

RTL Design Sherpa

RTL AMBA Wishbone B4 Module Reference — Rev 0.91

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1 Wishbone B4 Modules

Location: rtl/amba/wb4/ **Test Location:** val/amba/ **Status:** New (2026-09-09) — master, slave, monitor, retry, clock-gated and CDC variants, with sim and formal collateral. Family FULL sweep on one pinned seed base (RDS_SEED_BASE=20260910): 379 cells across the nine wb4 suites in val/amba/, plus the two AXI4-Lite bridge suites in projects/components/utility-ip/converters/dv/tests/, all passing. The cell count grew on 2026-09-11, when USE_BURST_HINTS became a dimension of the master and monitor suites.

1.1 Overview

A Wishbone B4 master and slave pair — and a monitor that can watch either — speaking the **pipelined** mode the B4 revision added, behind the same FUB-side contract the APB4 pair uses: a command queue and a response queue, each a valid/ready handshake through a gaxi_skid_buffer. A FUB that already talks to apb4_master through cmd_*/rsp_* talks to wb4_master the same way. What changes is the field names, and a 2-bit status in place of pslverr.

There is no Wishbone B5. B4 (2010) is the current revision of the specification, and that is what the wb4 name records — the way apb4 records AMBA 4 APB.

1.1.1 Protocol Scope

Signal set	Master (m_wb_*)	Slave (s_wb_*)
CYC, STB, WE, ADR, DAT_W (DAT_O), SEL	drives	receives
STALL	receives	drives (combinational from state, never from STB)
ACK, ERR, RTY, DAT_R (DAT_I)	receives	drives (registered)

- **Pipelined by default, classic by parameter.** With CLASSIC=0, a new STB every clock while STALL is low, and in-order termination. With CLASSIC=1 the blocks speak B4 standard ("classic") mode: the master holds the request on STB/CYC until the termination and ignores STALL; the slave never drives STALL, accepts a presentation once, and refuses an accept in the clock its termination is on the wire — the held STB is still there, and would otherwise be taken twice.

- **Match the mode to the peer.** The two modes do not mix, in either direction: a pipelined master drops `STB` after one clock, which a classic slave never accepts, and a classic master's held `STB` is accepted again every clock by a pipelined slave (B4 chapter 5). Both tests and both formal harnesses run each mode against a peer of the same mode.
- **RTY is a status, not a retry.** Every termination comes back to the FUB as `rsp_status` = ACK (0), ERR (1) or RTY (2). The master does not re-issue on RTY; the FUB decides, or `wb4_retry` decides for it. The encoding is `wb4_pkg`.
- **Burst hints are carried, not acted on.** `CTI` and `BTE` are advisory in B4. With `USE_BURST_HINTS = 1` the master puts the FUB's hint on the wires with the transfer it belongs to, and the slave hands the received hint to its FUB; neither changes behaviour, because deciding what a burst means is the peripheral's job. With the parameter at 0 the ports still exist and the bus reads CLASSIC/LINEAR — a legal non-burst cycle, which is what every consumer had before the hints landed.
- **Still not implemented, deliberately:** `LOCK` and the `TGA`/`TGC`/`TGD` tag signals.

1.2 Related Modules

- **wb4_master** — command/response queues in, pipelined Wishbone out; issue is credit-gated by the response queue so a termination can always be enqueued
- **wb4_slave** — pipelined Wishbone in, command/response queues out; in-order registered termination with an orphan-response guard
- **wb4_monitor** — watches either block's queues and reports completions, errors, timeouts, latency and address-range hits as monitor bus packets tagged `PROTOCOL_WB`; in-order tracking queue, since B4 terminates in issue order
- **wb4_retry** — RTY retry on the FUB side of the master: in-order completion buffer, re-issue up to a budget with a delay, the FUB sees RTY only once the budget is spent; **wb4_master_retry** is the block and the master together
- **wb4_master_cg**, **wb4_slave_cg** — the clock-gated variants (`amba_clock_gate_ctrl`, runtime enable and idle count, output valids masked for the wake overlap)
- **wb4_slave_cdc** — the slave with its queues carried to another clock domain over two `gaxi_fifo_async` instances; **wb4_slave_cdc_cg** adds the gated bus clock

- **wb4_master_stub** / **wb4_slave_stub** — the pair with packed cmd/rsp vectors, for shim-style converters
- **wb4_pkg** — the response-status encoding shared by the family and the DV

1.3 Design Notes

The ports are `clk` and `aresetn` (active-low) — the repository convention — rather than Wishbone's `CLK_I`/`RST_I` (active-high). An integrator bridging to a Wishbone fabric that carries `RST_I` inverts it at the boundary; the reset macros inside the blocks are polarity-agnostic.

1.4 Testing

Nine test files in `val/amba/`, all through the RDS-DV framework's Wishbone B4 BFMs (`CocoTBFramework.components.wb4`: `WB4Master`, `WB4Slave`, `WB4Monitor`) and the GAXI BFMs on the FUB-side queues:

- `test_wb4_master.py` — the master alone: the framework slave answers on the bus from a memory model with ERR and RTY address windows, the monitor checks the wires, and every response on `rsp_*` is paired in order with the termination the slave produced.
- `test_wb4_slave.py` — the slave alone: the framework master drives the bus and the test is the FUB. Includes the abort case (CYC dropped with requests outstanding), which a master-slave loop can never produce, and proves the slave discards the late responses instead of pairing them with the next cycle.
- `test_wb4_master_slave_loop.py` — both back to back (`rtl/amba/testcode/wb4_master_slave_loop.sv`); the framework monitor and the wrapper's own protocol checks must agree.
- `test_wb4_master_cg.py`, `test_wb4_slave_cg.py` — the clock-gated wrappers: the gate must not swallow a transfer, and the wake must be bounded.
- `test_wb4_slave_cdc.py` — the clock-domain-crossing slave, and the gated variant built on it.
- `test_wb4_master_retry.py` — the RTY retry wrapper: re-issue budget, delay, and that a retried transfer carries its own payload and burst hint.
- `test_wb4_monitor.py` — the monitor, decoded through the shared monbus path.
- `test_wb4_stubs.py` — the packed-vector stubs.

The three single-block tests above each assert the peak in-flight count exceeds one, so the pipelined mode is proven exercised rather than assumed. Formal covers nine blocks: seven under `formal/amba/wb4_*` and the two converter cores under `formal/converters/`.

1.5 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 “WISHBONE Classic Bus Cycles” (§3.1.3 handshaking protocol, §3.1.5 ACK/ERR/RTY termination, §3.2.2 and §3.2.4 classic pipelined single read and write with STALL), ch. 4 “WISHBONE Registered Feedback Bus Cycles” (CTI/BTE, §4.3), ch. 5 “Interfacing standard and pipelined peripherals”.
- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. “Theory of Latency-Insensitive Design.” *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 — the valid/ready channel discipline behind the `gaxi_skid_buffer` queues: a buffered stage may stall or add latency without changing what the endpoints compute.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 22 “Buses”, §22.1 “Bus Basics” and §22.2 “Bus Arbitration” — textbook treatment of the shared-bus model (secondary source).
- Chipressian. *Crack the Hardware Interview - from RTL Designers’ Perspective*, Part 2 ch. 2 “Handshake Protocols” (valid-ready protocol and valid-ready slices) — textbook treatment of the skid buffer (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `LatencyInsensitiveDesign__Carloni2001.pdf`)

2 wb4_master

2.1 Overview

wb4_master turns a command stream into Wishbone B4 **pipelined** bus cycles and returns every termination as a response. It is the Wishbone counterpart of apb4_master: the same cmd_*/rsp_* valid/ready queues on the FUB side, each a gaxi_skid_buffer. There is no state machine in here. STB follows the head of the command queue, the slave accepts a request on STB && !STALL, and every ACK, ERR or RTY is enqueued as it arrives. Terminations arrive in issue order — a B4 rule — so nothing is tagged.

Protocol scope: B4 pipelined. RTY is reported in rsp_status; the master never retries on its own. CTI/BTE burst hints are carried when USE_BURST_HINTS = 1 and tied to CLASSIC/LINEAR otherwise; no LOCK or tag signals.

2.2 Parameters

Parameter	Type	Default	Description
ADDR_WIDTH	int	32	Wishbone address width
DATA_WIDTH	int	32	Wishbone data width (port size)
CMD_DEPTH	int	4	Command queue depth in **entries**, 2..8
RSP_DEPTH	int	4	Response queue depth in **entries**, 2..8; also the maximum transfers in flight
CLASSIC	int	0	0 = B4 pipelined; 1 = B4 standard (“classic”) mode, see the family README. Match the peer: the modes do not mix
USE_BURST_HINTS	int	0	0 = CTI/BTE are driven CLASSIC/LINEAR and cmd_cti/cmd_bte are ignored; 1 = the FUB’s hints ride with their transfer onto the bus

Parameter	Type	Default	Description
SEL_WIDTH	int	DATA_WIDTH/8	Byte-select width (derived)

RSP_DEPTH is the outstanding limit. Wishbone gives a master no way to refuse a termination, so a request is only put on the bus when its response slot is already reserved. `RSP_DEPTH` therefore bounds both the queue and the pipeline depth; 4 sustains one transfer per clock against a slave with up to four clocks of termination latency.

2.3 Ports

The full declaration, straight from the RTL:

```
module wb4_master
    import wb4_pkg::*;
#(
    parameter int ADDR_WIDTH = 32,
    parameter int DATA_WIDTH = 32,
    parameter int CMD_DEPTH  = 4,
    parameter int RSP_DEPTH  = 4,
    parameter int CLASSIC    = 0,
    // Registered-feedback burst hints (B4 ch.4). 0 = the hint ports
read
    // CLASSIC/LINEAR and cmd_cti/cmd_bte are ignored.
    parameter int USE_BURST_HINTS = 0,
    parameter int SEL_WIDTH  = DATA_WIDTH / 8,
    // Short Parameters
    parameter int AW  = ADDR_WIDTH,
    parameter int DW  = DATA_WIDTH,
    parameter int SW  = SEL_WIDTH,
    parameter int STW = WB4_STATUS_WIDTH,
    parameter int CTW = WB4_CTI_WIDTH,
    parameter int BTW = WB4_BTE_WIDTH,
    // command packet: {we, adr, dat, sel} plus the hints when carried
    parameter int CPW = 1 + AW + DW + SW + ((USE_BURST_HINTS != 0) ?
CTW + BTW : 0),
    parameter int RPW = STW + DW            // response packet:
{status, dat}
) (
    input  logic              clk,
    input  logic              aresetn,

    // Wishbone B4 pipelined master
    output logic              m_wb_CYC,
    output logic              m_wb_STB,
    output logic              m_wb_WE,
    output logic [AW-1:0]     m_wb_ADR,
```

```
    output logic [DW-1:0]     m_wb_DAT_W,
    output logic [SW-1:0]     m_wb_SEL,
    output logic [CTW-1:0]    m_wb_CTI,      // burst hint; CLASSIC
when USE_BURST_HINTS=0
    output logic [BTW-1:0]    m_wb_BTE,      // burst type; LINEAR
when USE_BURST_HINTS=0
    input  logic              m_wb_STALL,
    input  logic              m_wb_ACK,
    input  logic              m_wb_ERR,
    input  logic              m_wb_RTY,
    input  logic [DW-1:0]     m_wb_DAT_R,

    // Command queue (FUB -> bus)
    input  logic              cmd_valid,
    output logic              cmd_ready,
    input  logic              cmd_we,
    input  logic [AW-1:0]     cmd_adr,
    input  logic [DW-1:0]     cmd_dat,
    input  logic [SW-1:0]     cmd_sel,
    input  logic [CTW-1:0]    cmd_cti,       // ignored when
USE_BURST_HINTS=0
    input  logic [BTW-1:0]    cmd_bte,       // ignored when
USE_BURST_HINTS=0

    // Response queue (bus -> FUB)
    output logic              rsp_valid,
    input  logic              rsp_ready,
    output logic [STW-1:0]    rsp_status,
    output logic [DW-1:0]     rsp_dat
);
```

2.3.1 Clock and Reset

Port	Width	Direction	Description
clk	1	Input	Bus clock
aresetn	1	Input	Active-low asynchronous reset (invert Wishbone `RST_I` at the boundary)

2.3.2 Wishbone Master Interface

Port	Width	Direction	Description
m_wb_CYC	1	Output	Cycle: high from the first `STB` until the last outstanding transfer

Port	Width	Direction	Description
			terminates
m_wb_STB	1	Output	Strobe: a request is presented
m_wb_WE	1	Output	Write enable
m_wb_ADR	ADDR_WIDTH	Output	Address
m_wb_DAT_W	DATA_WIDTH	Output	Write data (`DAT_O` in the specification)
m_wb_SEL	SEL_WIDTH	Output	Byte select, reads and writes
m_wb_CTI	3	Output	Burst hint, cycle type; CLASSIC unless `USE_BURST_HINTS=1`
m_wb_BTE	2	Output	Burst hint, burst type; LINEAR unless `USE_BURST_HINTS=1`
m_wb_STALL	1	Input	Slave cannot accept this clock; the request is held
m_wb_ACK	1	Input	Normal termination
m_wb_ERR	1	Input	Error termination
m_wb_RTY	1	Input	Retry termination
m_wb_DAT_R	DATA_WIDTH	Input	Read data (`DAT_I`)

2.3.3 Command Interface

Port	Width	Direction	Description
cmd_valid	1	Input	Command valid
cmd_ready	1	Output	Command queue has room
cmd_we	1	Input	Write (1) or read (0)
cmd_adr	ADDR_WIDTH	Input	Address
cmd_dat	DATA_WIDT	Input	Write data

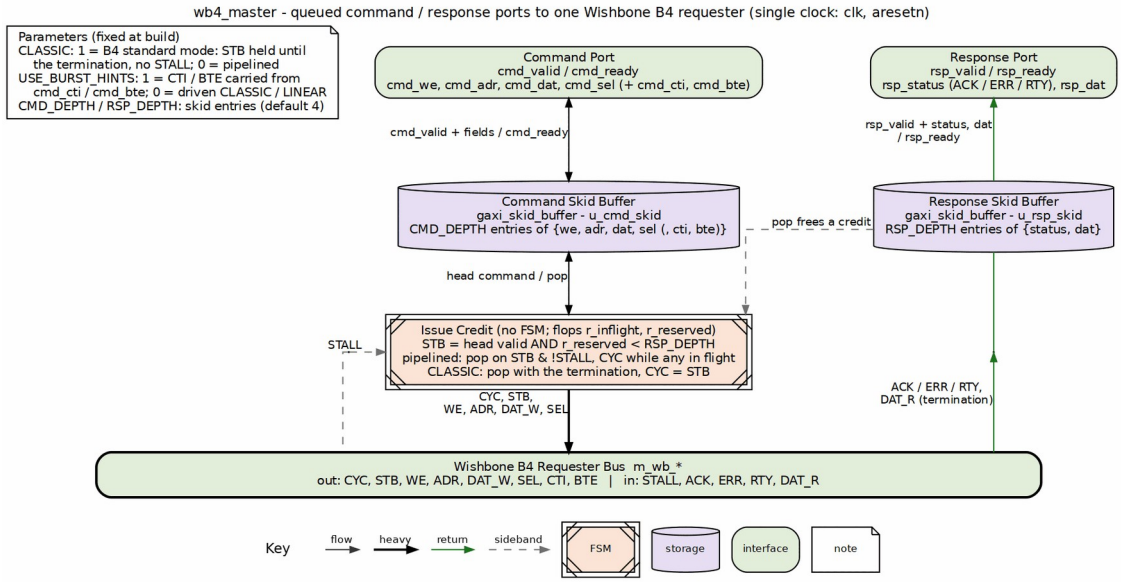
Port	Width	Direction	Description
	H		
cmd_sel	SEL_WIDTH	Input	Byte select
cmd_cti	3	Input	Burst hint for this transfer; ignored unless `USE_BURST_HINTS=1`
cmd_bte	2	Input	Burst type for this transfer; ignored unless `USE_BURST_HINTS=1`

2.3.4 Response Interface

Port	Width	Direction	Description
rsp_valid	1	Output	Response valid
rsp_ready	1	Input	FUB takes the response
rsp_status	2	Output	`WB4_RSP_ACK` (0), `WB4_RSP_ERR` (1), `WB4_RSP_RTY` (2)
rsp_dat	DATA_WIDTH	Output	Read data (undefined for writes, ERR and RTY)

2.4 Functional Description

2.4.1 Block diagram



wb4_master block diagram

Commands drop down the left lane: the FUB's command port feeds the command skid buffer, and the issue-credit logic puts the head on the bus as CYC/STB with its address, data and select. Terminations climb the right lane: ACK, ERR or RTY with DAT_R is pushed into the response skid and handed back on the response port. There is no FSM. Two counters gate issue (`r_reserved < RSP_DEPTH`), so a response slot is reserved before a transfer reaches the bus; Wishbone gives a master no way to refuse a termination. `CLASSIC` selects B4 standard mode, where STB is held until the termination and STALL is ignored.

Two counters and three wires do the work an APB master needs an FSM for:

```
issue    = cmd head valid && r_reserved < RSP_DEPTH
STB      = issue
CYC      = issue || r_inflight != 0
accept   = STB && !STALL          -> pop command, r_inflight++,
r_reserved++
term     = CYC && (ACK|ERR|RTY)   -> push {status, DAT_R},
r_inflight--
rsp pop  = rsp_valid && rsp_ready -> r_reserved--
```

`r_inflight` counts transfers the slave has accepted but not terminated; `r_reserved` counts everything issued that the FUB has not yet taken off the response queue — which is exactly what can occupy that queue in the worst case. Reserving at issue, rather than reading the queue's count back, avoids the "count is stale by one" reasoning the APB master's back-to-back path needs.

Classic mode (CLASSIC=1): the command retires with its termination (`accept = term`), so nothing is in flight between clocks; CYC follows STB; STALL is ignored. The credit gate still guarantees the push at termination has room, since `r_reserved` then counts only queued responses.

ERR and RTY are mutually exclusive in the specification. If a slave asserts both, ERR wins — a broken slave reads as an error rather than a retry.

2.5 Timing

Path	Clocks
`cmd_valid` accepted to STB	1 (command skid)
ACK/ERR/RTY to `rsp_valid`	1 (response skid)
Sustained throughput	one request and one termination per clock

A stalled request is held unchanged until accepted: the command skid's head does not move until `STB && !STALL`.

2.6 Usage Example

```
wb4_master #(
    .ADDR_WIDTH (32),
    .DATA_WIDTH (32),
    .CMD_DEPTH  (4),
    .RSP_DEPTH  (4)
) u_wb_master (
    .clk        (clk),
    .aresetn    (aresetn),
    .m_wb_CYC   (wb_cyc),   .m_wb_STB   (wb_stb),
    .m_wb_WE    (wb_we),    .m_wb_ADR   (wb_adr),
    .m_wb_DAT_W (wb_dat_o), .m_wb_SEL   (wb_sel),
    .m_wb_STALL (wb_stall), .m_wb_ACK   (wb_ack),
    .m_wb_ERR   (wb_err),   .m_wb_RTY   (wb_rty),
    .m_wb_DAT_R (wb_dat_i),
    .cmd_valid  (cmd_valid), .cmd_ready (cmd_ready),
    .cmd_we     (cmd_we),    .cmd_adr   (cmd_adr),
    .cmd_dat    (cmd_dat),   .cmd_sel   (cmd_sel),
    .rsp_valid  (rsp_valid), .rsp_ready (rsp_ready),
    .rsp_status (rsp_status),.rsp_dat   (rsp_dat)
);
```

2.7 Design Notes

- A termination that arrives with nothing in flight is a slave protocol violation. It is still enqueued (the FUB sees it) and reported in simulation.

- **Burst hints are carried, never acted on.** `CTI` and `BTE` are advisory in B4: the master puts the FUB's hint on the wires with the transfer it belongs to and changes nothing else. They ride inside the command queue, so a hint cannot slip onto a neighbouring transfer when the queue delays one. With `USE_BURST_HINTS = 0` the ports still exist and the bus reads CLASSIC/LINEAR, which is a legal non-burst Wishbone cycle.
- Under `ifdef FORMAL` the block asserts: `STB` implies `CYC`; `CYC` covers every in-flight transfer; a stalled request is held stable; the credit invariant `r_reserved <= RSP_DEPTH`; and that every termination found queue space.

2.8 Related Modules

- wb4_slave — the mirror image
- apb4_master — the same FUB-side contract over APB
- gaxi_skid_buffer — both queues

2.9 Testing

`val/amba/test_wb4_master.py` drives the block alone against the framework's Wishbone BFMs (`CocoTBFramework.components.wb4`), with the GAXI BFMs on the queues; `val/amba/test_wb4_master_slave_loop.py` runs it back to back with its counterpart. See the family README. Formal: `formal/amba/wb4_master/`.

2.10 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 "WISHBONE Classic Bus Cycles" (§3.1.3 handshaking protocol, §3.1.5 `ACK`/`ERR`/`RTY` termination, §3.2.2 and §3.2.4 classic pipelined single read and write with `STALL`), ch. 4 "WISHBONE Registered Feedback Bus Cycles" (`CTI`/`BTE`, §4.3), ch. 5 "Interfacing standard and pipelined peripherals".
- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. "Theory of Latency-Insensitive Design." *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 — the valid/ready channel discipline behind the `gaxi_skid_buffer` queues: a buffered stage may stall or add latency without changing what the endpoints compute.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 22 "Buses", §22.1 "Bus Basics" and §22.2 "Bus

Arbitration” — textbook treatment of the shared-bus model (secondary source).

- Chipressian. *Crack the Hardware Interview - from RTL Designers' Perspective*, Part 2 ch. 2 “Handshake Protocols” (valid-ready protocol and valid-ready slices) — textbook treatment of the skid buffer (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `LatencyInsensitiveDesign__Carloni2001.pdf`)

3 wb4_master_cg

3.1 Overview

The **clock-gated variant** of wb4_master: an `amba_clock_gate_ctrl` instance produces a gated clock that feeds an otherwise unmodified `wb4_master`. Functionally identical to the base module; gating is enabled and tuned by input signals, not parameters, and `cfg_cg_enable = 0` gives the base behaviour exactly. See the Shared book's Clock-Gated Variants Guide and amba_clock_gate_ctrl for the architecture and the gate cell.

3.2 Parameters

In addition to all `wb4_master` parameters:

Parameter	Type	Default	Description
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown; bounds the programmable idle threshold

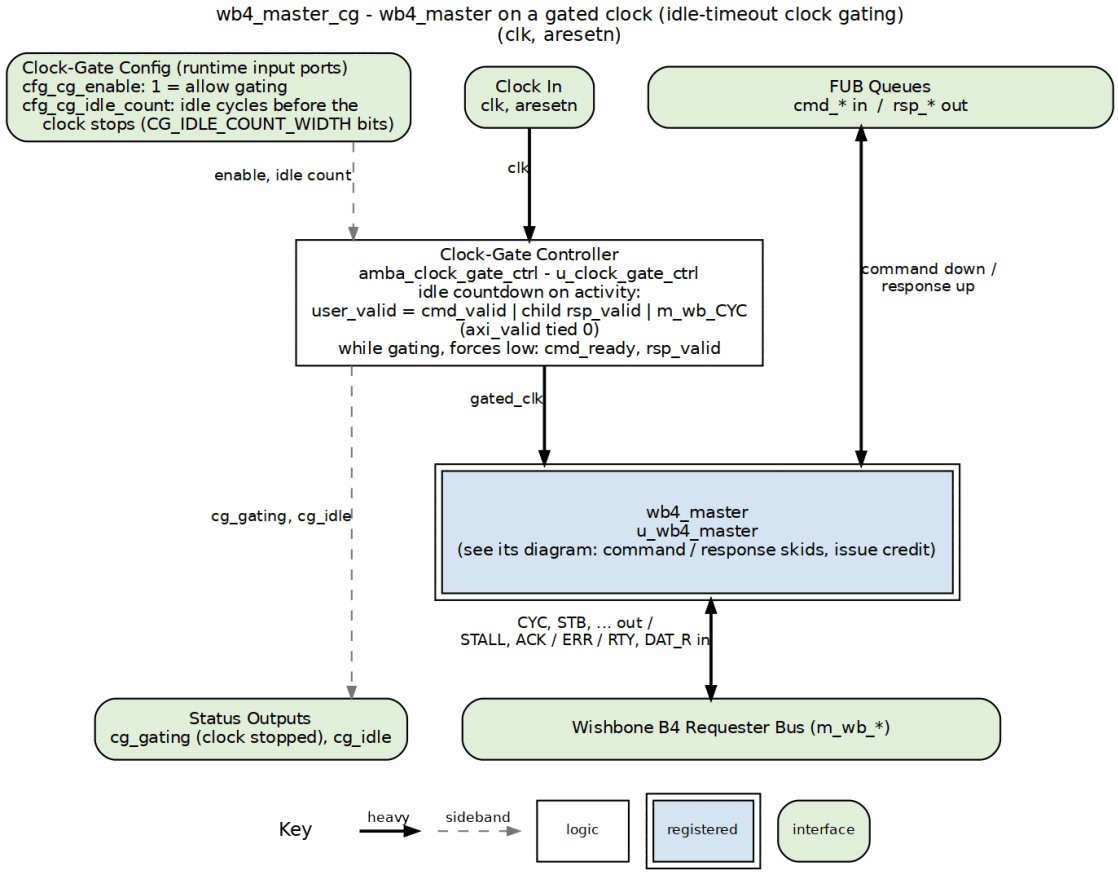
3.3 Ports

`wb4_master`'s ports plus:

Port	Direction	Description
`cfg_cg_enable`	in	Global clock-gate enable
`cfg_cg_idle_count [CG_IDLE_COUNT_WIDTH-1:0]`	in	Idle clocks before the clock is gated
`cg_gating`	out	Clock is gated now
`cg_idle`	out	Nothing pending (the controller's idle indicator)

3.4 Functional Description

3.4.1 Block diagram



wb4_master_cg block diagram

`wb4_master` is one child box (see its diagram); this wrapper adds only the clock gate. The heavy clock column runs from `clk` through `amba_clock_gate_ctrl` to the child's `gated_clk`. The controller wakes on `cmd_valid`, the child's `rsp_valid` or an open `m_wb_CYC`, and while the clock is stopped it forces `cmd_ready` and `rsp_valid` low so no handshake completes against a sleeping core.

3.4.2 Wake-Up Terms

The registered wake term follows the family rule — peer VALIDs and every place work can be pending, never a peer READY:

Term	Why
`cmd_valid`	a FUB command offered
response pending on the output	a termination waiting for the

Term	Why
	FUB
`m_wb_CYC`	a bus cycle open: requests queued, terminations due

`rsp_ready` is deliberately absent: a FUB that parks its ready high while idle would otherwise hold the clock on forever.

3.4.3 Masks for the Wake-Latency Overlap

Gating can engage on the same edge pending work appears, and the wake takes a clock or two. Two FUB-facing signals are therefore masked while `cg_gating` is high:

Signal	Masked to	Why
`rsp_valid` (to the FUB)	low	a FUB must not have a response retired by a stopped clock
`cmd_ready` (to the FUB)	low	a FUB must not have a command taken by a stopped clock

The masks only defer; once the pending term is high, gating cannot engage, so a visible valid is never truncated. The bus side needs no mask: `CYC`/`STB` come from registered state that is idle whenever the clock may stop, and a slave terminates only inside a cycle.

3.5 Design Notes

- Formal: `formal/amba/wb4_master_cg/` proves the wrapper's glue contract with a clock-enable model of the gate cell (a derived clock is not provable in the repo's single-clock flow; the stopped clock itself is the cocotb test's job): reset state, no gating while a cycle is open on the bus or a response is visible, no gating with the enable low, `cmd_ready` low while gated, and a command offered wakes the clock by its third clock; covers gating, an ungated transfer, a response handed back and re-gating.
- The wake term is combinational into the controller (which registers it once), not a second local flop as in `apb4_master_cg`; see `formal/amba/apb4_slave_cg/KNOWN_BUG.md` for the latency that flop adds.

3.6 Related Modules

- wb4_master, wb4_slave_cg

- apb4_master_cg — the same wrapper over APB

3.7 Testing

`val/amba/test_wb4_master_cg.py` runs the `wb4_master` phases with gating enabled and checks, every clock, that the clock is never gated while a cycle is open, that the clock does gate after each phase drains, and that it never gates with the enable low.

3.8 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 "WISHBONE Classic Bus Cycles", the protocol of the wrapped Wishbone block.
- Benini, L., Bogliolo, A., De Micheli, G. "A Survey of Design Techniques for System-Level Dynamic Power Management." *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 — the fixed-timeout policy `amba_clock_gate_ctrl` implements: gate after `cfg_cg_idle_count` idle cycles, wake on activity.
- Benini, L., De Micheli, G. "Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines." *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 — the latch-based gated clock (`icg`) at the bottom of the gating chain.
- Roy, S. *Advanced Digital System Design*, Springer, ch. 15 "Low-Power Digital System Design", §15.5.5 "Logic Level Power Down" — textbook treatment of clock gating (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `rtl/common/references/DynamicPowerManagementTimeout__Benini2000.pdf`, `rtl/common/references/LatchBasedClockGating__Benini1996.pdf`)

4 wb4_master_retry

4.1 Overview

`wb4_master_retry` is wb4_retry in front of wb4_master: the master's command/response queue contract and Wishbone B4 ports, with RTY terminations retried up to `cfg_max_retries` times, `cfg_retry_delay` clocks apart, before the FUB sees one. Drop it in where `wb4_master` would go when the slave is known to retry.

4.2 Parameters

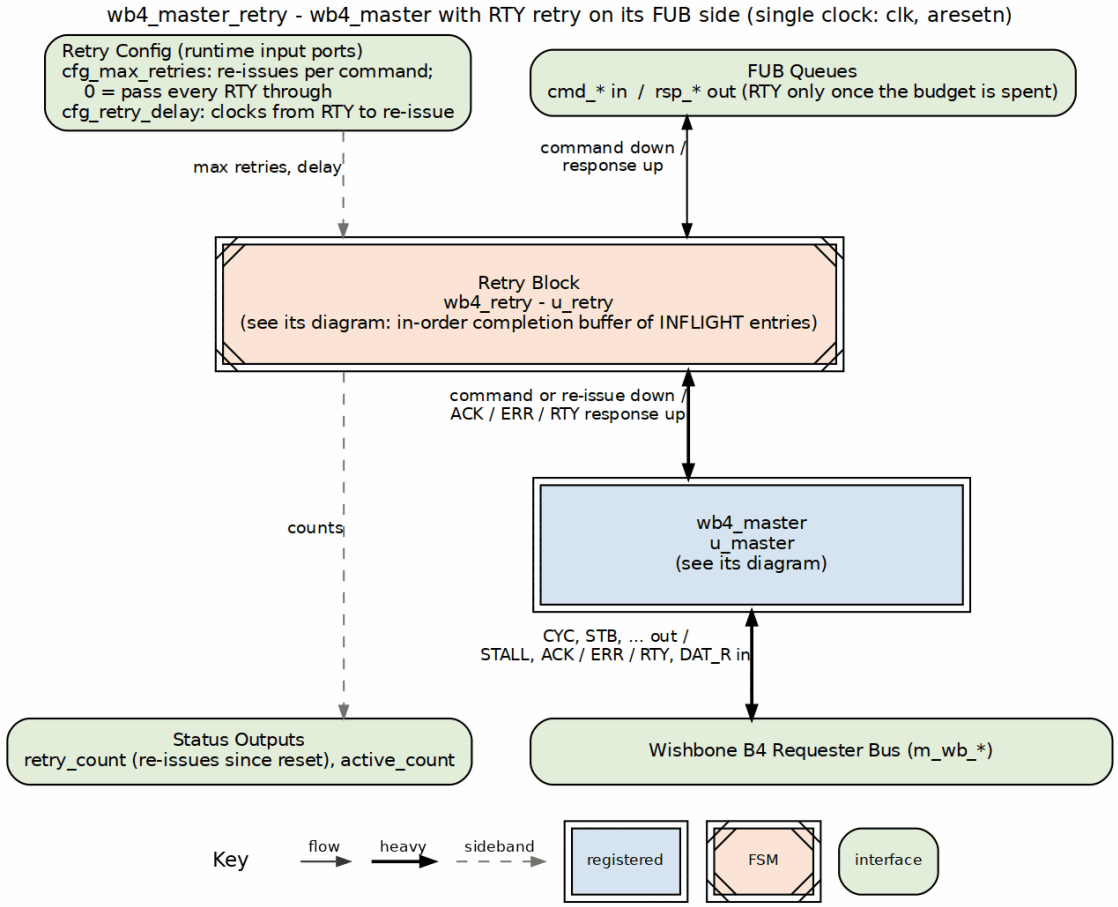
Parameter	Type	Default	Description
ADDR_WIDTH	int	32	Wishbone address width
DATA_WIDTH	int	32	Wishbone data width
CMD_DEPTH	int	4	Master command queue depth
RSP_DEPTH	int	4	Master response queue depth; bounds the transfers on the bus
CLASSIC	int	0	0 = B4 pipelined, 1 = B4 standard (classic) mode
INFLIGHT	int	1	Retry block completion-buffer depth. 1 = strict program order (see wb4_retry)
USE_BURST_HINTS	int	0	0 = `m_wb_CTI`/`m_wb_BTE` read CLASSIC/LINEAR and `cmd_cti`/`cmd_bte` are ignored; 1 = the hints are carried, and they ride **in the retry buffer** so a re-issued transfer carries its own hint

With `INFLIGHT = 1` the bus carries one transfer at a time regardless of `RSP_DEPTH`; set `INFLIGHT` up to `RSP_DEPTH` for pipelined traffic that tolerates a retried transfer completing behind later ones.

4.3 Ports

wb4_master's ports (clk, aresetn, m_wb_* including m_wb_CTI [2:0] and m_wb_BTE [1:0], cmd_* including cmd_cti [2:0] and cmd_bte [1:0], rsp_*) plus cfg_max_retries [7:0], cfg_retry_delay [15:0], retry_count [31:0] and active_count [7:0] from the retry block.

4.4 Block Diagram



wb4_master_retry block diagram

wb4_retry and wb4_master are one child box each (see their diagrams). FUB commands pass through the retry block, which re-issues RTY terminations up to cfg_max_retries times, cfg_retry_delay clocks apart, before the FUB sees one; the master drives the Wishbone bus. retry_count and active_count come from the retry block.

4.5 Usage Example

```
wb4_master_retry #(
    .ADDR_WIDTH (32),
```

```
    .DATA_WIDTH (32),
    .CMD_DEPTH  (4),
    .RSP_DEPTH  (4),
    .INFLIGHT   (1)
) u_wb_master (
    .clk (clk), .aresetn (rst_n),
    .cfg_max_retries (8'd3),
    .cfg_retry_delay (16'd16),
    .m_wb_CYC (wb_cyc), .m_wb_STB (wb_stb), .m_wb_WE
(wb_we), .m_wb_ADR (wb_adr),
    .m_wb_DAT_W (wb_dat_w), .m_wb_SEL (wb_sel),
    .m_wb_STALL (wb_stall), .m_wb_ACK (wb_ack), .m_wb_ERR
(wb_err), .m_wb_RTY (wb_rty),
    .m_wb_DAT_R (wb_dat_r),
    .cmd_valid (cmd_valid), .cmd_ready (cmd_ready), .cmd_we
(cmd_we), .cmd_adr (cmd_adr),
    .cmd_dat (cmd_dat), .cmd_sel (cmd_sel),
    .rsp_valid (rsp_valid), .rsp_ready (rsp_ready), .rsp_status
(rsp_status), .rsp_dat (rsp_dat),
    .retry_count (), .active_count ()
);
```

4.6 Related Modules

- wb4_retry, wb4_master

4.7 Testing

`val/amba/test_wb4_master_retry.py`; see wb4_retry.

4.8 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 – §3.1.5 “Use of [ACK_O], [ERR_O] and [RTY_O]”: the retry termination the wrapped `wb4_retry` re-issues. The retry budget and the in-order completion buffer are repo design with no paper behind them.
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`)

5 wb4_master_stub

5.1 Overview

wb4_master with its FUB-side queues presented as a single **packed vector** instead of named fields, matching what `apb4_master_stub` and `apb5_master_stub` offer. A shim-style converter that already carries a packed command through its own pipeline connects one bus instead of six, and the packing lives here rather than being re-derived by every consumer.

Nothing else differs: the stub instantiates the ordinary block and only splices the vector.

5.2 Parameters

Every parameter of wb4_master, plus the derived widths:

Parameter	Value	Description
CPW	1 + AW + DW + SW + 3 + 2	Packed command width
RPW	2 + DW	Packed response width

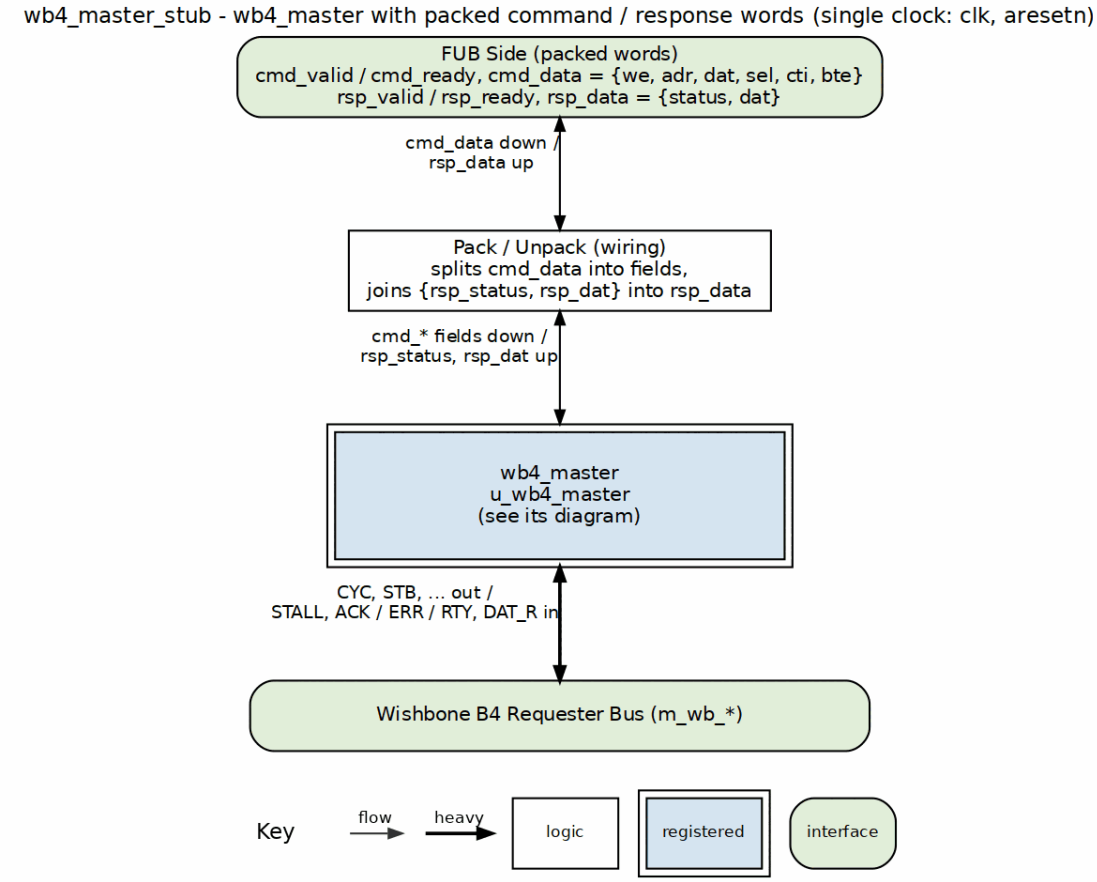
5.3 Ports

`wb4_master`'s Wishbone side, unchanged. On the FUB side, the command arrives packed and the response leaves packed:

Port	Direction	Description
`cmd_valid` / `cmd_ready`	Input / Output	Command handshake
`cmd_data [CPW-1:0]`	Input	Packed command
`rsp_valid` / `rsp_ready`	Output / Input	Response handshake
`rsp_data [RPW-1:0]`	Output	Packed response

5.4 Functional Description

5.4.1 Block diagram



wb4_master_stub block diagram

A packing wrapper: `wb4_master` is one child box (see its diagram). The FUB side carries one packed command word and one packed response word; the pack/unpack stage is wiring only, splitting `cmd_data` into its fields and joining `{rsp_status, rsp_dat}` into `rsp_data`.

5.4.2 Packing

Most significant field first. Both stubs use the same order, so a `wb4_master_stub` and a `wb4_slave_stub` connect directly:

```
cmd_data = {we, adr, dat, sel, cti, bte}      // CPW = 1 + AW + DW + SW
+ 3 + 2
rsp_data = {status, dat}                      // RPW = 2 + DW
```

`cti` and `bte` hold their bits whatever `USE_BURST_HINTS` is, so the vector width does not move with the parameter. With the hints off the inner block ignores them and the bus reads CLASSIC/LINEAR.

5.5 Related Modules

- wb4_master — the block this wraps
- wb4_slave_stub — the other half of a stub pair
- wb4_pkg — the status encoding inside `rsp_data`

5.6 Testing

`val/amba/test_wb4_stubs.py` checks the packing directly, in both directions and with the burst hints on and off. Two mutations fail it: swapping `dat`/`sel` in the master's unpack, and reversing the slave's response unpack.

5.7 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 "WISHBONE Classic Bus Cycles", the protocol of the wrapped Wishbone block.
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`)

6 wb4_monitor

6.1 Overview

wb4_monitor watches the cmd_*/rsp_* queues of a wb4_master or wb4_slave and reports what happens on the Wishbone side as monitor bus packets: a completion or an error per transfer, a timeout when a request or a termination is late, a latency figure when a transfer crosses a threshold, optional queue-activity debug events, and address-range violations through the shared apb_monitor_addr_check. It is the Wishbone member of the monitor family and has the same configuration, monitor bus and status ports as apb4_monitor; every packet is tagged PROTOCOL_WB (4'h5) and its event codes come from ocelus_wb4_pkg.

The queues are the timing-convenient proxy for the bus — the same place apb4_monitor attaches: a command handshake is a request the master will put on the bus, a response handshake is the termination the slave returned. One difference from APB drives the whole design. Wishbone B4 pipelined mode keeps several transfers open, and terminates them **in issue order**. The monitor therefore tracks transfers in an in-order queue, not a slot table. A slot table that pairs a response with "the first active slot" mis-pairs as soon as a freed slot is reused while an older one is still open — and pipelined traffic does that constantly. The in-order queue pairs the response with the oldest open request by construction.

6.2 Parameters

Parameter	Type	Default	Description
USE_MONITOR	bit	1	0 removes the body and ties the outputs to idle
N_ADDR_RANGES	int	0	Address-range checker ranges; 0 leaves the checker out
ADDR_WIDTH	int	32	Wishbone address width
DATA_WIDTH	int	32	Wishbone data width
UNIT_ID	logic [7:0]	8'h01	Unit id stamped into every packet
AGENT_ID	logic [15:0]	16'h000B	Agent id stamped into every packet
MAX_TRANSACTIONS	int	8	Depth of the in-order tracking queue; transfers open at once beyond it are reported, not tracked
MONITOR_FIFO_DEP	int	8	Event FIFO depth

Parameter	Type	Default	Description
TH			between event selection and the monitor bus
USE_BURST_HINTS	int	0	1 reports the transfer's CTI in aux_data[7:5]; 0 leaves those bits zero and ignores cmd_cti

MAX_TRANSACTIONS should match the RSP_DEPTH of the wb4_master (or the MAX_OUTSTANDING of the wb4_slave) it watches; those bound how many transfers can be open, so a queue of the same depth never overflows.

6.3 Ports

```
module wb4_monitor
    import ocelus_common_pkg::*;   // PROTOCOL_WB, PktType*, packet
builder
    import ocelus_wb4_pkg::*;      // WB_ERR_*, WB_TIMEOUT_*,
WB_COMPL_*, WB_PERF_*, WB_DEBUG_*
    import wb4_pkg::*;              // WB4_RSP_* status encoding
#(
    parameter bit USE_MONITOR         = 1'b1,
    parameter int N_ADDR_RANGES       = 0,
    parameter int ADDR_WIDTH          = 32,
    parameter int DATA_WIDTH          = 32,
    parameter logic [7:0]  UNIT_ID    = 8'h01,
    parameter logic [15:0] AGENT_ID   = 16'h000B,
    parameter int MAX_TRANSACTIONS    = 8,
    parameter int MONITOR_FIFO_DEPTH  = 8,
    parameter int USE_BURST_HINTS     = 0
)
(
    input  logic                      aclk,
    input  logic                      aresetn,

    // Command queue being watched
    input  logic                      cmd_valid,
    input  logic                      cmd_ready,
    input  logic                      cmd_we,
    input  logic [AW-1:0]             cmd_adr,
    input  logic [DW-1:0]             cmd_dat,
    input  logic [SW-1:0]             cmd_sel,
    input  logic [CTW-1:0]            cmd_cti,   // ignored when
USE_BURST_HINTS=0
```

```
    // Response queue being watched
    input  logic                        rsp_valid,
    input  logic                        rsp_ready,
    input  logic [1:0]                  rsp_status,     // WB4_RSP_ACK /
ERR / RTY
    input  logic [DW-1:0]               rsp_dat,

    // Configuration (same set as apb4_monitor)
    input  logic                        cfg_error_enable,
    input  logic                        cfg_timeout_enable,
    input  logic                        cfg_protocol_enable,
    input  logic                        cfg_slverr_enable,
    input  logic                        cfg_perf_enable,
    input  logic                        cfg_latency_enable,
    input  logic                        cfg_throughput_enable,
    input  logic                        cfg_debug_enable,
    input  logic                        cfg_trans_debug_enable,
    input  logic [3:0]                  cfg_debug_level,
    input  logic [15:0]                 cfg_cmd_timeout_cnt,
    input  logic [15:0]                 cfg_rsp_timeout_cnt,
    input  logic [31:0]                 cfg_latency_threshold,
    input  logic [15:0]                 cfg_throughput_threshold,

    // Address-range checker configuration (N_ADDR_RANGES > 0)
    input  logic                        cfg_addr_check_enable,
    input  logic [N-1:0]                cfg_addr_range_enable,
    input  logic [N-1:0][AW-1:0]        cfg_addr_range_low,
    input  logic [N-1:0][AW-1:0]        cfg_addr_range_high,

    input  monbus_timestamp_t           i_mon_time,

    // Monitor bus
    output logic                        monbus_valid,
    input  logic                        monbus_ready,
    output monitor_packet_t             monbus_packet,
    output monbus_timestamp_t           monbus_timestamp,

    // Status
    output logic [7:0]                  active_count,
    output logic [15:0]                 error_count,
    output logic [31:0]                 transaction_count
);
```

6.3.1 Command and Response Queues

The monitor is a pure observer: it samples `cmd_valid && cmd_ready` and `rsp_valid && rsp_ready` and never drives either queue. `cmd_dat` is accepted for port uniformity and not used; `rsp_dat` is reported only in the orphan-response packet.

6.3.2 Configuration

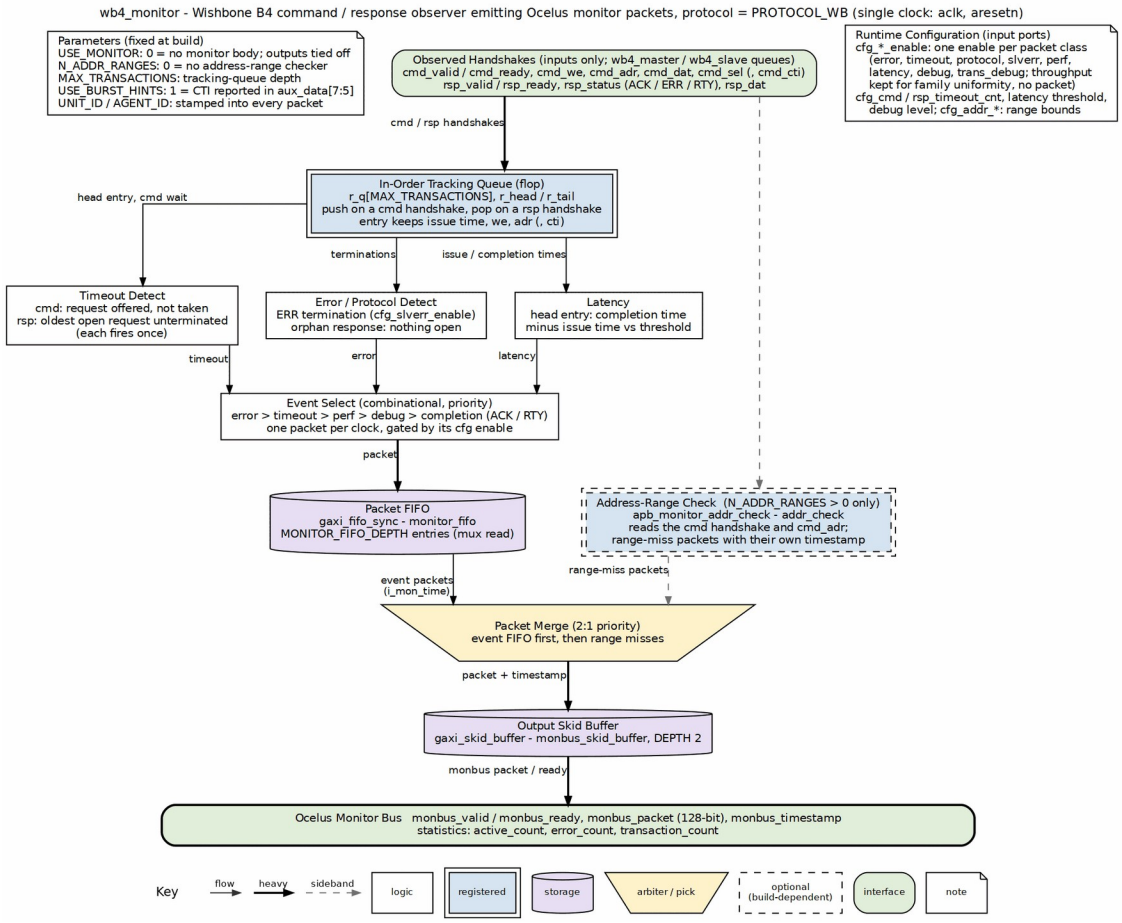
Signal	Effect
`cfg_error_enable`	Master enable for error packets
`cfg_slverr_enable`	Report an `ERR` termination as `WB_ERR_ERR`. This gates the `error_count` increment too: with it clear, an `ERR` termination is neither reported nor counted
`cfg_protocol_enable`	Report a response with nothing outstanding as `WB_ERR_ORPHAN_RSP`
`cfg_timeout_enable`	Enable both timeouts; a count of 0 disables that timeout
`cfg_cmd_timeout_cnt`	Clocks a request may sit on `cmd_valid` without `cmd_ready` before `WB_TIMEOUT_CMD`
`cfg_rsp_timeout_cnt`	Clocks the oldest open transfer may wait for its termination before `WB_TIMEOUT_RSP`
`cfg_perf_enable`, `cfg_latency_enable`	Emit a latency packet when a transfer's latency exceeds `cfg_latency_threshold`
`cfg_debug_enable`, `cfg_trans_debug_enable`	Emit `WB_DEBUG_QUEUE_ACTIVE` / `WB_DEBUG_QUEUE_IDLE` on the tracking queue's edges
`cfg_throughput_*`, `cfg_debug_level`	Accepted for family uniformity; no packet today

6.3.3 Status

`active_count` is the number of tracked transfers open right now, `transaction_count` the number terminated since reset, and `error_count` the number of `ERR` terminations (when `cfg_slverr_enable`), orphan responses and tracking overflows.

6.4 Functional Description

6.4.1 Block diagram



wb4_monitor block diagram

Same family shape as `apb4_monitor`. The command and response handshakes feed an in-order tracking queue: Wishbone terminates in issue order, so each response pairs with the oldest open request. The timeout, error/protocol and latency detectors feed a priority event select, which writes at most one packet per clock into the packet FIFO. An optional address-range checker (dashed, `N_ADDR_RANGES > 0`) produces its own range-miss packets. A 2:1 priority merge (the trapezium) feeds the output skid buffer onto the Ocelus monitor bus.

6.4.2 Transaction Tracking

A command handshake pushes `{we, adr, sel[3:0], cti, timestamp}` at the tail of the queue; a response handshake pops the head. The head is the transfer the response belongs to, because B4 terminates in issue order. Two edge cases are reported rather than guessed at:

- **Orphan response.** A response handshake with an empty queue. With `cfg_protocol_enable` it is a `WB_ERR_ORPHAN_RSP` packet carrying `rsp_dat` and the status; `transaction_count` does not move.
- **Tracking lost.** A command handshake with the queue full. The transfer is not tracked (its later response will show up as an orphan) and a `WB_ERR_TRACK_LOST` packet carries its address. Size `MAX_TRANSACTIONS` to the master's `RSP_DEPTH` and this never happens.

6.4.3 Event Detection

Event	Packet type	Event code	event_data[31:0]	aux (event_data[39:32])
Termination `ACK`	Completion	`WB_COMPL_READ` / `WB_COMPL_WRITE`	address	`{cti[2:0], sel[3:0], we}`
Termination `RTY`	Completion	`WB_COMPL_RTY`	address	`{cti[2:0], sel[3:0], we}`
Termination `ERR`	Error	`WB_ERR_ERR`	address	`{cti[2:0], sel[3:0], we}`
Response, nothing open	Error	`WB_ERR_ORPHAN_RSP`	`rsp_dat`	`{6'b0, status}`
Command, queue full	Error	`WB_ERR_TRACK_LOST`	address	`{cti[2:0], sel[3:0], we}`
`cmd_valid` stalled	Timeout	`WB_TIMEOUT_CMD`	address on `cmd_adr`	stall count (low 8 bits)
Head transfer late	Timeout	`WB_TIMEOUT_RSP`	head address	age (low 8 bits)
Latency over threshold	Perf	`WB_PERF_READ_LATENCY` / `WB_PERF_WRITE_LATENCY`	latency in clocks	`{cti[2:0], sel[3:0], we}`
Queue non-empty / empty	Debug	`WB_DEBUG_QUEUE_ACTIVE` / `WB_DEBUG_QUEUE_IDLE`	occupancy	0
Address out of	Error	`WB_ERR_ADDR_`	see	see

Event	Packet type	Event code	event_data[31:0]	aux (event_data[39:32])
range		RANGE	apb_monitor_addr_check	apb_monitor_addr_check

cti[2:0] is zero on a USE_BURST_HINTS=0 build, which is what every consumer decoded before the hints existed, so the packet format did not change for anyone who does not ask for them. Only CTI is reported: the aux byte has exactly three spare bits, which is WB4_CTI_WIDTH, and BTE is deliberately left out rather than squeezed in. The hint travels **in the tracking entry**, so a completion reports the CTI of its own transfer even with several open at once — reading cmd_cti at completion time would report whatever request happened to be on the queue then.

RTY is a completion, not an error: the FUB decides whether to retry, and the monitor reports what the slave said. Each timeout fires **once**: the command timeout once per stall (it re-arms when the request is taken or withdrawn), the response timeout once per queue entry (a flag in the entry). When the head pops and the next entry is already older than the limit, that entry is reported on the following clock.

6.4.4 Monitor Packet Format

Standard 128-bit packet from create_monitor_packet with a 64-bit side-band timestamp taken from i_mon_time at emission:

- packet_type per the table above; protocol = PROTOCOL_WB (4'h5)
- event_code from ocelus_wb4_pkg; channel_id = 0 (Wishbone has no IDs)
- unit_id = UNIT_ID, agent_id = AGENT_ID
- event_data[63:40] = 0, [39:32] = aux, [31:0] = value per the table

6.4.5 Event Priority and Loss

One event is written to the FIFO per clock, in the order error, timeout, perf, debug, completion; a lower-priority event that lands in the same clock as a higher one is dropped, and so is any event that arrives while the FIFO is full. The tracking queue does not wait for the packet, so active_count and transaction_count stay exact. error_count is a single increment per clock, so two error events landing together — an ERR termination and a tracking overflow in the same clock, which pipelined traffic can reach — advance it by one rather than two. This is the family's lossy-but-honest contract; size MONITOR_FIFO_DEPTH and the monitor bus consumer for the burstiest traffic the design can produce.

The event FIFO is a gaxi_fifo_sync in mux-read mode (REGISTERED=0): rd_data is valid in the clock of the read handshake, which is when the packet is built. The registered-read mode presents rd_data one clock after the handshake (the framework's fifo_flop BFM contract) and, read combinationally, re-presents the popped entry for a clock on back-to-back reads — so a burst of events would duplicate one packet and lose the next. This module's first test caught exactly that; see TASK-086 for the siblings.

6.5 Timing

- Push and pop are registered from the queue handshakes; `active_count` updates on the clock after a handshake.
- A completion packet leaves the event FIFO one clock after the response handshake and the skid buffer one clock later, so `monbus_valid` for a termination rises two clocks after `rsp_valid && rsp_ready` when the bus is idle.
- `monbus_valid` is held until `monbus_ready` (skid buffer).

6.6 Usage Example

```
wb4_monitor #(
    .ADDR_WIDTH       (32),
    .DATA_WIDTH       (32),
    .UNIT_ID          (8'h02),
    .AGENT_ID         (16'h0020),
    .MAX_TRANSACTIONS (4),          // = wb4_master RSP_DEPTH
    .N_ADDR_RANGES    (2)
) u_wb_mon (
    .aclk (clk), .aresetn (rst_n),
    .cmd_valid (cmd_valid), .cmd_ready (cmd_ready), .cmd_we (cmd_we),
    .cmd_adr (cmd_adr), .cmd_dat (cmd_dat), .cmd_sel (cmd_sel),
    .rsp_valid (rsp_valid), .rsp_ready (rsp_ready),
    .rsp_status (rsp_status), .rsp_dat (rsp_dat),
    .cfg_error_enable (1'b1), .cfg_timeout_enable (1'b1),
    .cfg_protocol_enable (1'b1), .cfg_slverr_enable (1'b1),
    .cfg_perf_enable (1'b0), .cfg_latency_enable (1'b0),
    .cfg_throughput_enable (1'b0), .cfg_debug_enable (1'b0),
    .cfg_trans_debug_enable (1'b0), .cfg_debug_level (4'h0),
    .cfg_cmd_timeout_cnt (16'd256), .cfg_rsp_timeout_cnt (16'd1024),
    .cfg_latency_threshold (32'd0), .cfg_throughput_threshold (16'd0),
    .cfg_addr_check_enable (1'b1), .cfg_addr_range_enable (2'b11),
    .cfg_addr_range_low ('{32'h0000_0000, 32'h4000_0000}),
    .cfg_addr_range_high ('{32'h0FFF_FFFF, 32'h4FFF_FFFF}),
    .i_mon_time (mon_time),
    .monbus_valid (monbus_valid), .monbus_ready (monbus_ready),
    .monbus_packet (monbus_packet), .monbus_timestamp
(monbus_timestamp),
    .active_count (), .error_count (), .transaction_count ()
);
```

6.7 Design Notes

- Under `ifdef FORMAL` the block asserts the queue occupancy bound, that a pop only happens with something open, and that an orphan is only flagged

when nothing is. `formal/amba/wb4_monitor/` adds the port-level properties (reset, protocol tag, valid-held, occupancy tracks the handshakes, `transaction_count` moves only on a pop) and covers every packet class.

- `USE_MONITOR=0` leaves the ports in place and ties `monbus_valid` and the counters to zero, so a build can drop the monitor without touching the wrapper.
- `ocelus_wb4_pkg` is imported by this module only; it is deliberately not re-exported by `ocelus_pkg`, so no existing consumer's filelist changes. `PROTOCOL_WB` itself lives in `ocelus_common_pkg` (an additive enum entry).

6.8 Related Modules

- wb4_master, wb4_slave — what it watches
- apb4_monitor — the family template
- apb_monitor_addr_check — the shared range checker, tagged `PROTOCOL_WB` here through its `PROTOCOL` parameter
- monitor_package_spec — packet format and protocol ids

6.9 Testing

`val/amba/test_wb4_monitor.py` drives both queues with GAXI BFMs (a producer/consumer pair on each side, an age-based responder) and decodes the packets through the shared `MonbusSlave/TBClasses.monbus.parse` path. Phases: ACK/ERR/RTY mix in command order, several transfers open (in-order pairing), latency threshold, both timeouts once each, orphan, tracking overflow, debug edges, and address range on the `N_ADDR_RANGES=2` build. Four RTL mutations (timeout re-fire, swapped read/write codes, pairing off by one, orphan not reported) all fail the test.

6.10 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — §3.1.5 (`ERR` and `RTY` terminations, reported as error and retry events) and ch. 4 (`CTI`/`BTE`, carried as burst hints); the timeouts are repo monitoring policy with no spec or paper behind them.
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`)

7 wb4_pkg

7.1 Overview

The response-status encoding shared by every module in the Wishbone B4 family and by the Python bus functional models, together with the registered-feedback burst hints of B4 chapter 4. Three enums and three width constants, kept in a package so each encoding has exactly one definition.

7.2 Functional Description

7.2.1 Declarations

```
package wb4_pkg;
    localparam int WB4_STATUS_WIDTH = 2;

    typedef enum logic [WB4_STATUS_WIDTH-1:0] {
        WB4_RSP_ACK = 2'b00,   // normal completion (read data valid)
        WB4_RSP_ERR = 2'b01,   // slave error
        WB4_RSP_RTY = 2'b10    // slave asks for a retry
    } wb4_status_t;

    // Registered-feedback burst hints (B4 chapter 4), advisory.
CLASSIC and
    // LINEAR are zero so a tied-off bus is a legal non-burst bus.
    localparam int WB4_CTI_WIDTH = 3;
    localparam int WB4_BTE_WIDTH = 2;

    typedef enum logic [WB4_CTI_WIDTH-1:0] {
        WB4_CTI_CLASSIC    = 3'b000,
        WB4_CTI_CONST_ADDR = 3'b001,
        WB4_CTI_INCR       = 3'b010,
        WB4_CTI_RSVD_3     = 3'b011,
        WB4_CTI_RSVD_4     = 3'b100,
        WB4_CTI_RSVD_5     = 3'b101,
        WB4_CTI_RSVD_6     = 3'b110,
        WB4_CTI_EOB        = 3'b111
    } wb4_cti_t;

    typedef enum logic [WB4_BTE_WIDTH-1:0] {
        WB4_BTE_LINEAR = 2'b00,
        WB4_BTE_WRAP4  = 2'b01,
        WB4_BTE_WRAP8  = 2'b10,
        WB4_BTE_WRAP16 = 2'b11
    } wb4_bte_t;
endpackage : wb4_pkg
```

Name	Value	Meaning
WB4_STATUS_WIDTH	2	Width of every rsp_status / cmd-side status field in the family
WB4_RSP_ACK	2'd0	Normal termination
WB4_RSP_ERR	2'd1	Slave signalled an error
WB4_RSP_RTY	2'd2	Slave asked for the transfer to be retried
(unused)	2'd3	Not an encoding; treated as reserved

RTY is a status, not an action. wb4_master never re-issues on its own; it reports what the slave said and the FUB decides. Put wb4_retry in front of the master to absorb retries transparently up to a budget.

7.2.2 Burst Hint Encodings

Registered-feedback cycles (B4 chapter 4). Both are **advisory**: the family carries them when USE_BURST_HINTS = 1 and neither the master nor the slave changes behaviour on them.

wb4_cti_t	Value	Meaning
WB4_CTI_CLASSIC	3'b000	Classic cycle, no burst
WB4_CTI_CONST_ADDR	3'b001	Constant-address burst (a FIFO-like target)
WB4_CTI_INCR	3'b010	Incrementing burst
WB4_CTI_RSVD_3 .. WB4_CTI_RSVD_6	3'b011-3'b110	Reserved by the spec
WB4_CTI_EOB	3'b111	End-of-burst: the last transfer of the run

wb4_bte_t	Value	Meaning
WB4_BTE_LINEAR	2'b00	Linear burst
WB4_BTE_WRAP4	2'b01	4-beat wrap
WB4_BTE_WRAP8	2'b10	8-beat wrap
WB4_BTE_WRAP16	2'b11	16-beat wrap

WB4_CTI_WIDTH is 3 and WB4_BTE_WIDTH is 2. Both encodings put the non-burst case at zero on purpose, so a bus with the hint wires tied off is a legal classic bus rather than an illegal encoding.

7.3 Design Notes

This is a package, not a module. No ports, no clock. Compile order matters: it must be analysed before anything importing it, which is why rtl/amba/filelists/wb4_pkg.f exists and every consumer -f includes it rather than hand-listing the source.

Why a status field at all. Wishbone terminates a transfer with one of three wires, ACK, ERR or RTY. The FUB-side response queue carries the outcome as a 2-bit status instead, so a consumer reads one field rather than decoding three mutually exclusive strobes. WB4_STATUS_WIDTH sizes that field everywhere it appears.

7.4 Related Modules

Every module in the family imports the package: wb4_master, wb4_slave, wb4_monitor, wb4_retry, wb4_master_retry, the clock-gated and CDC variants, and the axil4_to_wb4 bridge in projects/components/utility-ip/converters.

The DV side mirrors the same three values in CocoTBFramework.components.shared.wb4_common (WB4_STATUS_ACK, WB4_STATUS_ERR, WB4_STATUS_RTY), and ocelus_wb4_pkg carries the separate event codes the monitor emits.

- Wishbone B4 family README — the protocol scope and the mode rules
- ocelus_wb4_pkg — the monitor's event codes, a different package

7.5 Testing

No test of its own, and none expected. A package has no behaviour to simulate. It is verified by every module that imports it failing to elaborate if a declaration is wrong.

7.6 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 4 "WISHBONE Registered Feedback Bus Cycles", §4.3 "Signal Description": the CTI cycle-type and BTE burst-type encodings this package defines.
- Indexed in rtl/amba/references/README.md (local PDF: WishboneB4Spec__OpenCores2010.pdf)

8 wb4_retry

8.1 Overview

`wb4_retry` sits on the FUB side of `wb4_master` and turns a Wishbone RTY into a retry. Every command the FUB issues is held in an in-order completion buffer until its answer has been handed back. ACK and ERR complete the entry as they arrive. RTY puts the entry back on the issue path after `cfg_retry_delay` clocks, until `cfg_max_retries` re-issues have been spent; only then does the FUB see the RTY. Responses reach the FUB in command order whatever happened on the bus, and re-issues always take priority over new commands.

`cfg_max_retries = 0` makes the block a pass-through. `wb4_master_retry` is the block and the master together, with the master's ports.

Order on the bus. A retried command re-appears on the bus behind the commands issued after it, which with `INFLIGHT > 1` may already have terminated. The FUB still gets its responses in order, but a read issued behind a write that was retried can observe memory from before that write. `INFLIGHT = 1`, the default, keeps program order exactly: one command on the bus at a time, retries included. Use `INFLIGHT > 1` only for traffic with no ordering dependence across transfers.

8.2 Parameters

Parameter	Type	Default	Description
ADDR_WIDTH	int	32	Wishbone address width
DATA_WIDTH	int	32	Wishbone data width
INFLIGHT	int	1	Completion-buffer entries: commands accepted from the FUB and not yet answered. 1 = strict program order

8.3 Ports

```
module wb4_retry
    import wb4_pkg::*;
#(
    parameter int ADDR_WIDTH = 32,
    parameter int DATA_WIDTH = 32,
    parameter int INFLIGHT   = 1,
    // Short Parameters
    parameter int AW  = ADDR_WIDTH,
    parameter int DW  = DATA_WIDTH,
    parameter int SW  = DW/8,
```

```
    parameter int STW = WB4_STATUS_WIDTH,
    parameter int CTW = WB4_CTI_WIDTH,
    parameter int BTW = WB4_BTE_WIDTH
)
(
    input  logic              clk,
    input  logic              aresetn,

    input  logic [7:0]        cfg_max_retries,   // 0 = pass RTY
through
    input  logic [15:0]       cfg_retry_delay,   // clocks from RTY to
re-issue

    // FUB side: the same contract wb4_master offers
    input  logic              cmd_valid,
    output logic              cmd_ready,
    input  logic              cmd_we,
    input  logic [AW-1:0]     cmd_adr,
    input  logic [DW-1:0]     cmd_dat,
    input  logic [SW-1:0]     cmd_sel,
    input  logic [CTW-1:0]    cmd_cti,           // burst hint, stored
with its transfer
    input  logic [BTW-1:0]    cmd_bte,
    output logic              rsp_valid,
    input  logic              rsp_ready,
    output logic [1:0]        rsp_status,        // WB4_RSP_ACK /
ERR / RTY
    output logic [DW-1:0]     rsp_dat,

    // wb4_master side
    output logic              mst_cmd_valid,
    input  logic              mst_cmd_ready,
    output logic              mst_cmd_we,
    output logic [AW-1:0]     mst_cmd_adr,
    output logic [DW-1:0]     mst_cmd_dat,
    output logic [SW-1:0]     mst_cmd_sel,
    output logic [CTW-1:0]    mst_cmd_cti,
    output logic [BTW-1:0]    mst_cmd_bte,
    input  logic              mst_rsp_valid,
    output logic              mst_rsp_ready,
    input  logic [1:0]        mst_rsp_status,
    input  logic [DW-1:0]     mst_rsp_dat,

    output logic [31:0]       retry_count,       // re-issues since
reset
    output logic [7:0]        active_count       // entries in the
```

```
    buffer
);
```

8.3.1 Configuration

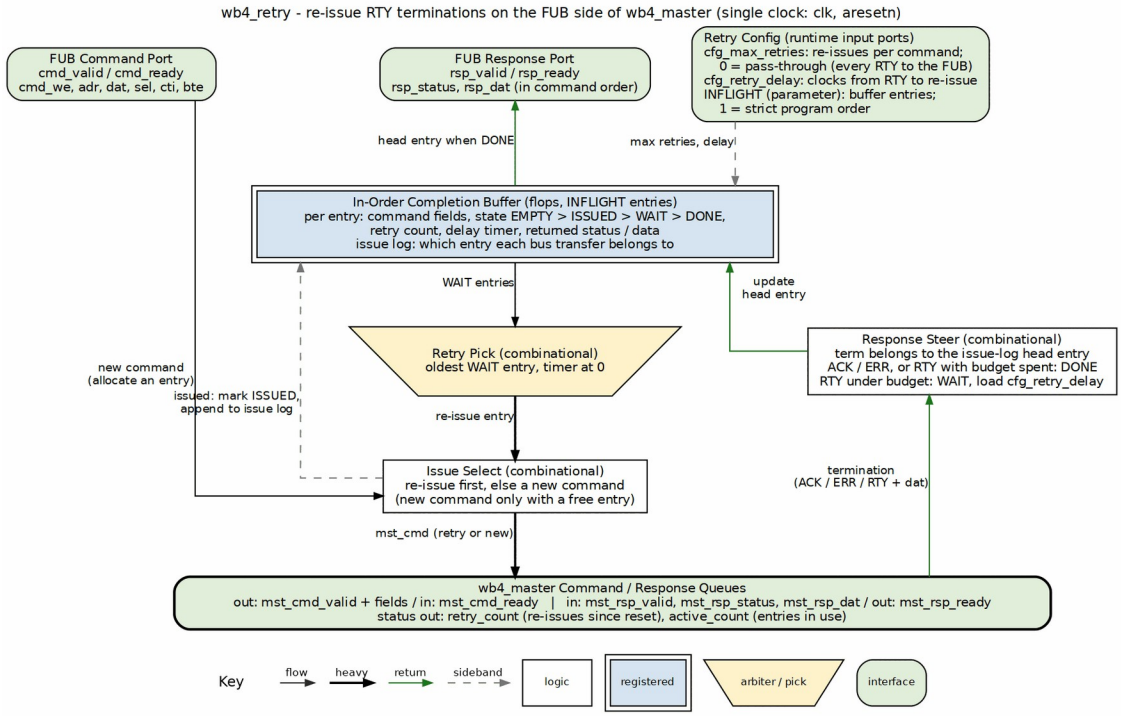
`cfg_max_retries` is the number of re-issues allowed per command, not the number of attempts: with 3, a command is presented to the bus at most four times. `cfg_retry_delay` is the number of clocks between the RTY and the re-issue; 0 re-issues on the next clock the master can take a command. Both are sampled when used, so they can change between transfers.

8.3.2 Status

`retry_count` counts every re-issue since reset. `active_count` is the number of commands accepted from the FUB and not yet answered, at most `INFLIGHT`.

8.4 Functional Description

8.4.1 Block diagram



wb4_retry block diagram

New FUB commands drop down the left lane into the issue select and are allocated an entry in the in-order completion buffer as they issue. The master's terminations climb the right lane into the response steer, which updates the issue-log head entry: ACK, ERR, or an RTY with the budget spent marks it DONE; an RTY under budget puts it back in WAIT with `cfg_retry_delay` loaded. The retry pick (the trapezium) chooses the oldest WAIT entry whose delay has run out, and a re-issue

takes priority over a new command. Responses leave from the buffer head only once it is DONE, so the FUB sees them in command order.

Each entry of the completion buffer holds the command (`we`, `adr`, `dat`, `sel`), a retry count, a delay timer and, once answered, the status and read data. Entries are allocated at the tail when a command is accepted and released at the head when the FUB takes the response, so the FUB order is the buffer order. A separate issue log records which entry each command handed to the master belongs to, in the order the master received them; Wishbone terminates in that order, so the log head is always the entry a termination belongs to.

Event	Effect
`cmd_valid && cmd_ready`	Entry allocated at the tail and forwarded to `mst_cmd_*` in the same clock
`mst_rsp_*` handshake, `ACK` or `ERR`	Log-head entry marked done with the status and data
`mst_rsp_*` handshake, `RTY`, retries left	Entry marked waiting, retry count +1, timer loaded with `cfg_retry_delay`, `retry_count` +1
`mst_rsp_*` handshake, `RTY`, budget spent	Entry marked done with `RTY`
Waiting entry with timer at 0	Re-issued on `mst_cmd_*` before any new command (oldest first)
Head entry done	`rsp_valid`; released on the handshake

`cmd_ready` is low while a retry is pending issue, while the buffer is full, and while the master cannot take a command — so a FUB never has a command accepted that would be reordered behind a retry. `mst_rsp_ready` is high whenever a command is on the bus: the termination's entry is waiting for it, so the master is never back-pressured on a response.

8.5 Timing

- A new command reaches `mst_cmd_*` combinationally in the clock it is accepted; the master's own skid buffer registers it.
- A termination is recorded in the clock it arrives; `rsp_valid` for the head follows one clock later (registered state).
- With `INFLIGHT = 1` the bus carries one transfer at a time, so the sustained rate is one transfer per round trip. With `INFLIGHT = N` and a master `RSP_DEPTH >= N`, up to N transfers are open.

8.6 Design Notes

- Under `ifdef FORMAL` the block asserts the occupancy bounds of both the buffer and the issue log, that everything on the bus is an allocated entry, that a retry is never issued in the clock a new command is accepted, and that the FUB only ever sees the head once it is done. `formal/amba/wb4_retry/` adds the port-level contract (payload equality of every re-issue, `1 + retries` issues per command, the budget and the delay, `RTY` to the FUB only once the budget is spent) at `INFLIGHT = 1` and the bounds at `INFLIGHT = 2`.
- The block does not know why the slave retried. Back-off is a fixed delay; a slave that needs a longer or growing gap needs a larger `cfg_retry_delay` from the controlling software.

8.7 Related Modules

- wb4_master_retry — this block with `wb4_master` behind it
- wb4_master — what it drives
- axil4_to_wb4 — a bridge that maps `RTY` to an AXI error; put this block between it and the master to retry instead

8.8 Testing

`val/amba/test_wb4_master_retry.py`, through the wrapper: the framework Wishbone slave answers with address windows that retry a bounded number of times, retry forever, or error, so every FUB response, the number of re-issues (`retry_count`, the monitor's transfer count, the slave's `RTY` count) and the read data follow from the model. Formal: `formal/amba/wb4_retry/`.

8.9 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — §3.1.5 "Use of [ACK_O], [ERR_O] and [RTY_O]": the retry termination this block re-issues. The retry budget and the in-order completion buffer are repo design with no paper behind them.
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`)

9 wb4_slave

9.1 Overview

`wb4_slave` accepts Wishbone B4 **pipelined** transfers one per clock and presents each on a command queue; it terminates them, in order, from a response queue with a registered ACK, ERR or RTY and read data. It is the Wishbone counterpart of `apb4_slave`, with the same `cmd_*`/`rsp_*` valid/ready contract on the FUB side. There is no state machine here either: `STALL` is the inverse of "can accept", and the termination side is one counter and one register.

Protocol scope: B4 pipelined. The FUB chooses the termination through `rsp_status`, so a FUB can answer RTY without the slave knowing why.

9.2 Parameters

Parameter	Type	Default	Description
ADDR_WIDTH	int	32	Wishbone address width
DATA_WIDTH	int	32	Wishbone data width (port size)
CMD_DEPTH	int	2	Command queue depth in **entries**, 2..8
RSP_DEPTH	int	2	Response queue depth in **entries**, 2..8
MAX_OUTSTANDING	int	16	Transfers accepted but not yet terminated before `STALL` asserts
CLASSIC	int	0	0 = B4 pipelined; 1 = B4 standard ("classic") mode, see the family README. Match the peer: the modes do not mix
USE_BURST_HINTS	int	0	0 = the bus `CTI`/`BTE` inputs are ignored and `cmd_cti`/`cmd_bte` read CLASSIC/LINEAR; 1 = each transfer's hints are handed to the FUB with

Parameter	Type	Default	Description
			it
SEL_WIDTH	int	DATA_WIDTH/8	Byte-select width (derived)

`MAX_OUTSTANDING` bounds the FUB's in-order pipeline, not the master's: a master's own limit is its response queue. Set it to at least the number of commands the FUB can hold before it produces the first response, or the bus stalls short of the FUB's throughput.

9.3 Ports

```
module wb4_slave
    import wb4_pkg::*;
#(
    parameter int ADDR_WIDTH      = 32,
    parameter int DATA_WIDTH      = 32,
    parameter int CMD_DEPTH       = 2,
    parameter int RSP_DEPTH       = 2,
    parameter int MAX_OUTSTANDING = 16,
    parameter int CLASSIC         = 0,
    parameter int SEL_WIDTH       = DATA_WIDTH / 8,
    // Short Parameters
    parameter int AW  = ADDR_WIDTH,
    parameter int DW  = DATA_WIDTH,
    parameter int SW  = SEL_WIDTH,
    parameter int STW = WB4_STATUS_WIDTH,
    parameter int CTW = WB4_CTI_WIDTH,
    parameter int BTW = WB4_BTE_WIDTH,
    // command packet: {we, adr, dat, sel} plus the hints when carried
    parameter int CPW = 1 + AW + DW + SW + ((USE_BURST_HINTS != 0) ?
CTW + BTW : 0),
    parameter int RPW = STW + DW            // response packet:
{status, dat}
) (
    input  logic              clk,
    input  logic              aresetn,

    // Wishbone B4 pipelined slave
    input  logic              s_wb_CYC,
    input  logic              s_wb_STB,
    input  logic              s_wb_WE,
    input  logic [AW-1:0]     s_wb_ADR,
    input  logic [DW-1:0]     s_wb_DAT_W,
    input  logic [SW-1:0]     s_wb_SEL,
    input  logic [CTW-1:0]    s_wb_CTI,      // burst hint; ignored
when USE_BURST_HINTS=0
```

```
    input  logic [BTW-1:0]    s_wb_BTE,
    output logic              s_wb_STALL,
    output logic              s_wb_ACK,
    output logic              s_wb_ERR,
    output logic              s_wb_RTY,
    output logic [DW-1:0]     s_wb_DAT_R,

    // Command queue (bus -> FUB)
    output logic              cmd_valid,
    input  logic              cmd_ready,
    output logic              cmd_we,
    output logic [AW-1:0]     cmd_adr,
    output logic [DW-1:0]     cmd_dat,
    output logic [SW-1:0]     cmd_sel,
    output logic [CTW-1:0]    cmd_cti,       // CLASSIC when
USE_BURST_HINTS=0
    output logic [BTW-1:0]    cmd_bte,       // LINEAR  when
USE_BURST_HINTS=0

    // Response queue (FUB -> bus)
    input  logic              rsp_valid,
    output logic              rsp_ready,
    input  logic [STW-1:0]    rsp_status,
    input  logic [DW-1:0]     rsp_dat
);
```

9.3.1 Clock and Reset

Port	Width	Direction	Description
clk	1	Input	Bus clock
aresetn	1	Input	Active-low asynchronous reset

9.3.2 Wishbone Slave Interface

Port	Width	Direction	Description
s_wb_CYC	1	Input	Cycle
s_wb_STB	1	Input	Strobe: a request is presented
s_wb_WE	1	Input	Write enable
s_wb_ADR	ADDR_WIDTH	Input	Address

Port	Width	Direction	Description
s_wb_DAT_W	DATA_WIDTH	Input	Write data (DAT_I at the slave)
s_wb_SEL	SEL_WIDTH	Input	Byte select
s_wb_CTI	3	Input	Burst hint, cycle type; ignored unless USE_BURST_HINTS=1
s_wb_BTE	2	Input	Burst hint, burst type; ignored unless USE_BURST_HINTS=1
s_wb_STALL	1	Output	Cannot accept this clock; combinational from queue room and the outstanding count, never from STB
s_wb_ACK	1	Output	Normal termination (registered, one clock)
s_wb_ERR	1	Output	Error termination (registered, one clock)
s_wb_RTY	1	Output	Retry termination (registered, one clock)
s_wb_DAT_R	DATA_WIDTH	Output	Read data (DAT_O at the slave); holds between terminations

9.3.3 Command Interface

Port	Width	Direction	Description
cmd_valid	1	Output	Command valid
cmd_ready	1	Input	FUB takes the command
cmd_we	1	Output	Write (1) or read (0)
cmd_adr	ADDR_WIDTH	Output	Address
cmd_dat	DATA_WIDTH	Output	Write data
cmd_sel	SEL_WIDTH	Output	Byte select
cmd_cti	3	Output	Burst hint off the bus;

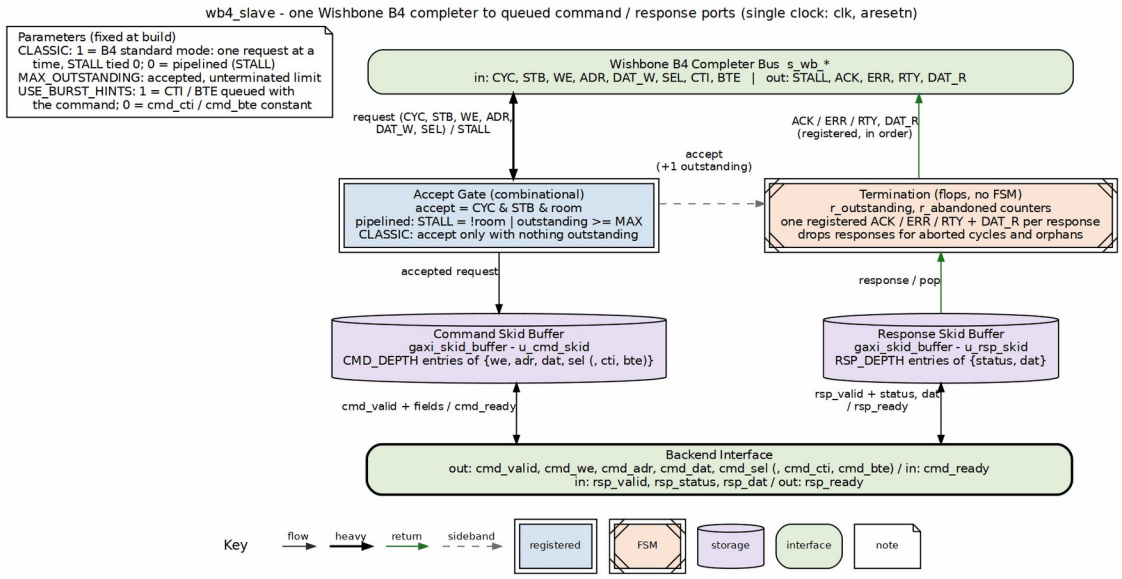
Port	Width	Direction	Description
			CLASSIC unless `USE_BURST_HINTS=1`
cmd_bte	2	Output	Burst type off the bus; LINEAR unless `USE_BURST_HINTS=1`

9.3.4 Response Interface

Port	Width	Direction	Description
rsp_valid	1	Input	Response valid
rsp_ready	1	Output	Response queue has room
rsp_status	2	Input	`WB4_RSP_ACK` (0), `WB4_RSP_ERR` (1), `WB4_RSP_RTY` (2)
rsp_dat	DATA_WIDTH	Input	Read data

9.4 Functional Description

9.4.1 Block diagram



wb4_slave block diagram

The Wishbone completer bus is at the top and the queued backend at the bottom. An accepted request (CYC, STB and room) drops down the left lane into the command skid; in pipelined mode

STALL is the only backpressure. Responses climb the right lane through the response skid into the termination logic, which drives exactly one registered ACK, ERR or RTY with DAT_R per response, in order. The outstanding and abandoned counters drop responses that belong to an aborted cycle, or that arrive with nothing open.

```
STALL  = !cmd queue room || r_outstanding == MAX_OUTSTANDING
accept = CYC && STB && !STALL     -> push {we, adr, dat, sel},
r_outstanding++
term   = rsp head valid && r_outstanding != 0 && CYC && r_abandoned ==
0
         -> register ACK|ERR|RTY from status, DAT_R, pop,
r_outstanding--
orphan = rsp head valid && (r_outstanding == 0 || r_abandoned != 0) ->
pop and drop
abort  = !CYC && r_outstanding != 0 -> r_abandoned += r_outstanding,
r_outstanding <= 0
```

Classic mode (`CLASSIC=1`): `STALL` is driven low; a request is accepted when nothing is outstanding and no termination is on the wire this clock, so a held presentation is taken exactly once; `MAX_OUTSTANDING` is effectively 1.

Orphan guard. A response with nothing outstanding cannot belong to any transfer (a duplicate from the FUB, or a response to a transfer the master abandoned). It is dropped, and reported in simulation. Left in the queue it would terminate the *next* transfer and every later response would be off by one: the positional mis-pairing `apb4_slave`'s guard exists for.

Abort. A master that drops `CYC` with transfers outstanding has ended the cycle. The outstanding count moves to an *abandoned* count, and that many later responses from the FUB are dropped as they arrive, even if the master has started a new cycle by then. Without that, a late response for an abandoned transfer would terminate the new cycle's first transfer — which is exactly what the slave test's abort phase caught before the count existed. Terminations are only ever driven inside a cycle, and never while a response is still owed to an abandoned transfer.

9.5 Timing

Path	Clocks
`STB` accepted to `cmd_valid`	1 (command skid)
`rsp_valid` to `ACK`/`ERR`/`RTY`	2 (response skid, then the output register)
Sustained throughput	one accept and one termination per clock

9.6 Usage Example

```
wb4_slave #(
    .ADDR_WIDTH      (32),
```

```
    .DATA_WIDTH      (32),
    .CMD_DEPTH       (2),
    .RSP_DEPTH       (2),
    .MAX_OUTSTANDING (16)
) u_wb_slave (
    .clk        (clk),
    .aresetn    (aresetn),
    .s_wb_CYC   (wb_cyc),   .s_wb_STB   (wb_stb),
    .s_wb_WE    (wb_we),    .s_wb_ADR   (wb_adr),
    .s_wb_DAT_W (wb_dat_i), .s_wb_SEL   (wb_sel),
    .s_wb_STALL (wb_stall), .s_wb_ACK   (wb_ack),
    .s_wb_ERR   (wb_err),   .s_wb_RTY   (wb_rty),
    .s_wb_DAT_R (wb_dat_o),
    .cmd_valid  (cmd_valid), .cmd_ready (cmd_ready),
    .cmd_we     (cmd_we),    .cmd_adr   (cmd_adr),
    .cmd_dat    (cmd_dat),   .cmd_sel   (cmd_sel),
    .rsp_valid  (rsp_valid), .rsp_ready (rsp_ready),
    .rsp_status (rsp_status),.rsp_dat   (rsp_dat)
);
```

9.7 Design Notes

- The FUB must answer commands in the order it received them; the slave pairs responses to transfers by position.
- Under `ifdef FORMAL` the block asserts: at most one of ACK/ERR/RTY per clock; a termination only for a transfer that was outstanding inside a cycle; `r_outstanding <= MAX_OUTSTANDING`; and that every accept found queue room.

9.8 Related Modules

- wb4_master — the mirror image
- apb4_slave — the same FUB-side contract over APB
- gaxi_skid_buffer — both queues

9.9 Testing

`val/amba/test_wb4_slave.py` drives the block alone against the framework's Wishbone BFMs (`CocoTBFramework.components.wb4`), with the GAXI BFMs on the queues; `val/amba/test_wb4_master_slave_loop.py` runs it back to back with its counterpart. See the family README. Formal: `formal/amba/wb4_slave/`.

9.10 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 “WISHBONE Classic Bus Cycles” (§3.1.3 handshaking protocol, §3.1.5 `ACK`/`ERR`/`RTY` termination, §3.2.2 and §3.2.4 classic pipelined single read and write with `STALL`), ch. 4 “WISHBONE Registered Feedback Bus Cycles” (`CTI`/`BTE`, §4.3), ch. 5 “Interfacing standard and pipelined peripherals”.
- Carloni, L.P., McMillan, K.L., Sangiovanni-Vincentelli, A.L. “Theory of Latency-Insensitive Design.” *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, 20(9), Sept. 2001. DOI: 10.1109/43.945302 — the valid/ready channel discipline behind the `gaxi_skid_buffer` queues: a buffered stage may stall or add latency without changing what the endpoints compute.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 22 “Buses”, §22.1 “Bus Basics” and §22.2 “Bus Arbitration” — textbook treatment of the shared-bus model (secondary source).
- Chipressian. *Crack the Hardware Interview - from RTL Designers’ Perspective*, Part 2 ch. 2 “Handshake Protocols” (valid-ready protocol and valid-ready slices) — textbook treatment of the skid buffer (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `LatencyInsensitiveDesign__Carloni2001.pdf`)

10 wb4_slave_cdc

10.1 Overview

wb4_slave with its command and response queues carried across a clock-domain boundary. The Wishbone bus lives in `wb_clk`; the FUB's `cmd_*`/`rsp_*` queues live in `aclk`. Two `gaxi_fifo_async` instances do the crossing (commands `wb_clk -> aclk`, responses `aclk -> wb_clk`), the same structure as apb4_slave_cdc. In-order termination is preserved because both crossings are FIFOs.

10.2 Parameters

In addition to all `wb4_slave` parameters:

Parameter	Type	Default	Description
CDC_DEPTH	int	4	Async FIFO depth, floored at 4; power of two preferred (Gray pointers)
USE_JOHNSON	int	0	0 = Gray pointers (power-of-two depth), 1 = Johnson pointers (any depth)

10.3 Ports

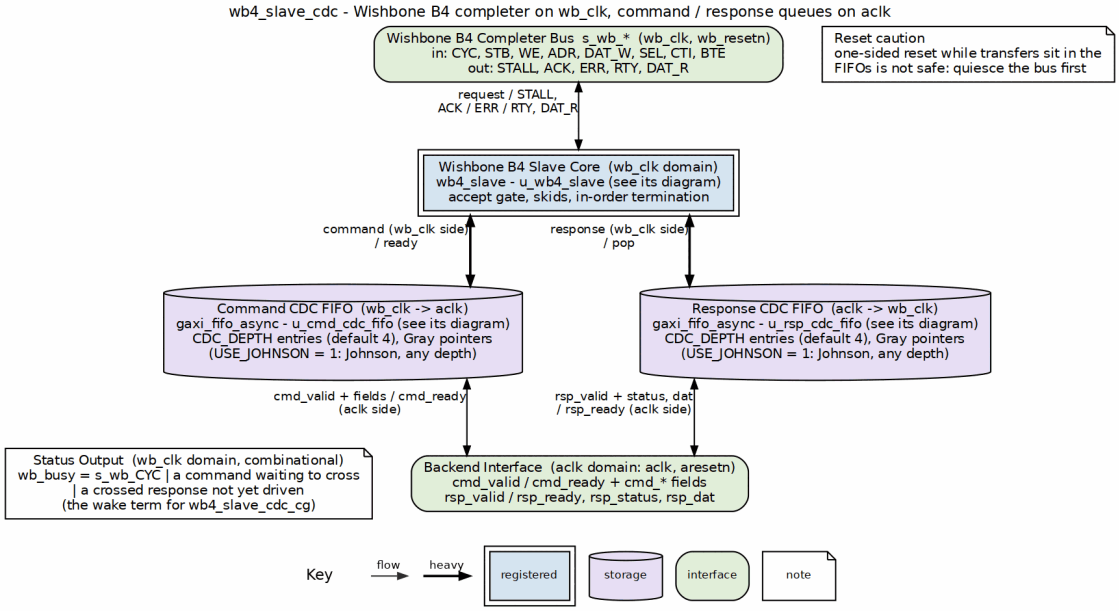
`wb4_slave`'s ports, with the clock and reset split by domain:

Port	Domain	Description
`wb_clk`, `wb_resetn`	Wishbone	the `s_wb_*` bus
`aclk`, `aresetn`	FUB	the `cmd_*` / `rsp_*` queues
`wb_busy` (output)	Wishbone	high while this block has anything in flight on the Wishbone side. Built from `wb_clk` signals ONLY, so a clock gate in that domain can use it directly; `wb4_slave_cdc_cg` is its

Port	Domain	Description
		one consumer

10.4 Functional Description

10.4.1 Block diagram



wb4_slave_cdc block diagram

The Wishbone domain (`wb_clk`) sits above the FUB domain (`aclk`). `wb4_slave` runs on `wb_clk`, and two `gaxi_fifo_async` children (see their diagram) straddle the boundary: commands cross down the left (`wb_clk` to `aclk`) and responses climb the right (`aclk` to `wb_clk`). Both are FIFOs, so the n-th response still terminates the n-th request. `wb_busy` is built from `wb_clk` signals only, which is what lets `wb4_slave_cdc_cg` use it as its wake term.

10.4.2 Clock Domains

The slave itself, its skid buffers, the `STALL` logic and the registered terminations are all in `wb_clk`. `MAX_OUTSTANDING` bounds the requests accepted while responses are pending, so the response FIFO can never overflow the slave's response queue whatever the clock ratio.

10.4.3 CDC Structure

```
 s_wb_* (wb_clk) --> wb4_slave --> cmd FIFO (wb_clk -> aclk) --> cmd_*
(aclk)
                                <-- rsp FIFO (aclk -> wb_clk) <-- rsp_*
(aclk)
```

Each FIFO is a `gaxi_fifo_async` with two-flop pointer synchronisers (`N_FLOP_CROSS = 2`). Throughput is one transfer per clock of the slower side once the pointers have crossed; latency is a few clocks of each domain per direction.

10.4.4 Reset Behavior

Each FIFO resets its own side's pointers from that side's reset. While a reset is asserted the resetting side is self-consistent, but a **one-sided reset with transfers in the FIFOs is not safe**: a write-side reset alone lets the read side see phantom occupancy (it fabricates entries), and a read-side reset alone replays consumed entries. Quiesce the bus before resetting one side. The full analysis is in apb4_slave_cdc, Reset Behavior, and applies here unchanged.

10.5 Related Modules

- wb4_slave, wb4_slave_cg
- gaxi_fifo_async — the crossing

10.6 Testing

`val/amba/test_wb4_slave_cdc.py` runs the `wb4_slave` phases with the Wishbone BFMs on `wb_clk` and the queue BFMs on `aclk` at several clock ratios in both directions (bus faster, bus slower). Formal: `formal/amba/wb4_slave_cdc/`, single-clock model (both domains on one clock) for the port-level contract, as the APB CDC harness does.

10.7 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 "WISHBONE Classic Bus Cycles", the protocol of the wrapped Wishbone block.
- Cummings, C.E. "Simulation and Synthesis Techniques for Asynchronous FIFO Design." *SNUG San Jose*, 2002 — the Gray-pointer async FIFO (`gaxi_fifo_async`) that carries the command and response across the two clocks.
- Ginosar, R. "Metastability and Synchronizers: A Tutorial." *IEEE Design & Test of Computers*, 28(5), Sept./Oct. 2011. DOI: 10.1109/MDT.2011.113 — synchronizer depth versus MTBF for the pointer crossings.
- Deschamps, J.-P., Sutter, G.D., Cantó, E. *Guide to FPGA Implementation of Arithmetic Functions*, Springer (LNEE 149), 2012, ch. 5 "Electronic Aspects of Digital Design", §5.3.5.3 "Asynchronous FIFO" — textbook treatment (secondary source).

- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `rtl/cdc/references/AsyncFifoGrayPointers__Cummings2002.pdf`, `rtl/cdc/references/MetastabilitySynchronizers__Ginosar2011.pdf`)

11 wb4_slave_cdc_cg

11.1 Overview

wb4_slave_cdc with the **Wishbone-side clock gated**, the combined variant matching `apb4_slave_cdc_cg` / `apb5_slave_cdc_cg`.

Only the Wishbone domain is gated. The FUB domain keeps running, so a consumer can go on draining commands and posting responses while the bus sleeps; the async FIFOs make that safe. Gating both would need a second controller and buys nothing a caller cannot do by gating its own clock.

11.2 Parameters

Every `wb4_slave_cdc` parameter, plus `CG_IDLE_COUNT_WIDTH`:

Parameter	Type	Default	Description
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown; bounds the programmable idle threshold

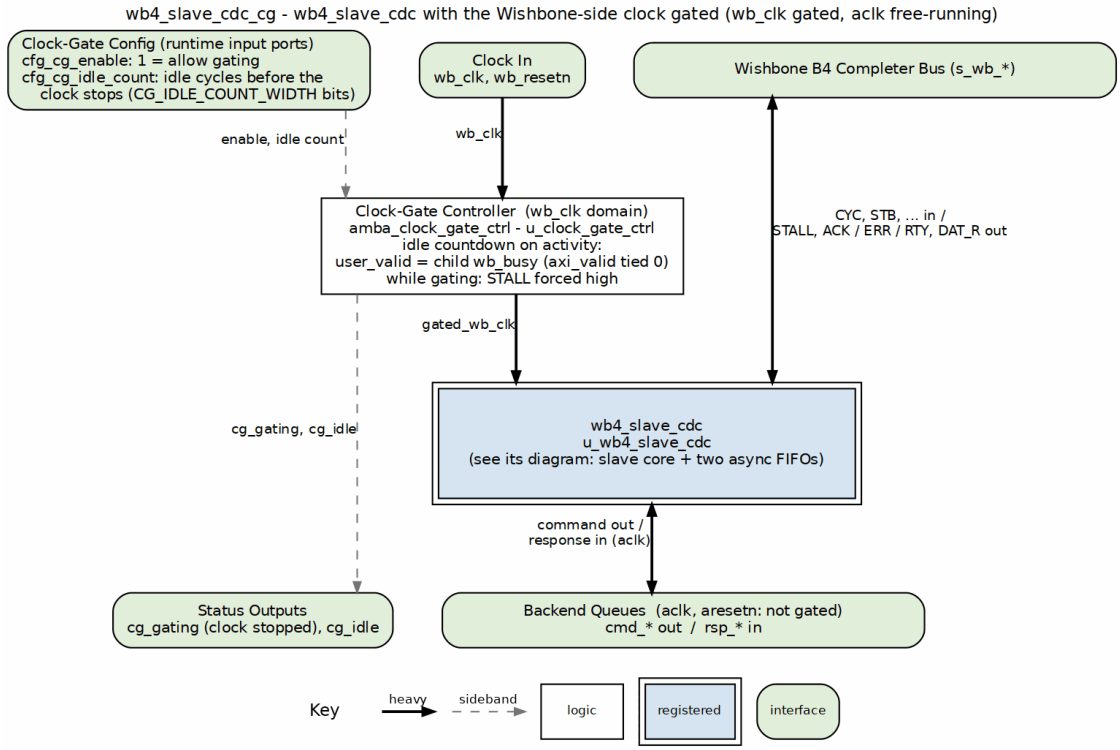
11.3 Ports

`wb4_slave_cdc`'s ports plus `cfg_cg_enable`, `cfg_cg_idle_count`, `cg_gating` and `cg_idle`, all in the `wb_clk` domain:

Port	Direction	Description
`cfg_cg_enable`	in	Global clock-gate enable
`cfg_cg_idle_count [CG_IDLE_COUNT_WIDTH-1:0]`	in	Idle clocks before the clock is gated
`cg_gating`	out	Clock is gated now
`cg_idle`	out	Nothing pending (the controller's idle indicator)

11.4 Functional Description

11.4.1 Block diagram



wb4_slave_cdc_cg block diagram

`wb4_slave_cdc` is one child box (see its diagram); this wrapper gates only the Wishbone clock. The controller wakes on the child's `wb_busy` and holds STALL high while the clock is stopped. The FUB domain (`aclk`) keeps running, so a consumer can drain commands and post responses while the bus sleeps; the async FIFOs inside the child make that safe.

11.4.2 The Wake Term Has to Be Single-Domain

This is the one thing worth reading before changing it. The wake term is `wb4_slave_cdc`'s `wb_busy` output, which is built **only** from `wb_clk` signals: a cycle open on the bus, a command waiting to cross, or a response that has crossed and not yet been driven.

Nothing from the `aclk` side may be used, because sampling it in `wb_clk` would be an unsynchronised crossing. That is the mistake this wrapper invites, and the reason `wb_busy` exists rather than the wake term being assembled here from whatever looked convenient.

For the same reason `cmd_valid` is **not** masked with `cg_gating` the way wb4_slave_cg masks it: `cmd_valid` lives in `aclk`, and masking it with a `wb_clk`-domain signal would be exactly that crossing. `s_wb_STALL` **is** held high while gated, for the reason `wb4_slave_cg` gives: a pipelined

accept is `STB && !STALL` in the master's clock, and a frozen "room" would let a master move on from a request the slave never sampled.

11.4.3 Reset Behavior

`wb4_slave_cdc`'s one-sided reset caveat applies unchanged: quiesce the bus before resetting one side.

11.5 Related Modules

- wb4_slave_cdc, wb4_slave_cg

11.6 Testing

`val/amba/test_wb4_slave_cdc.py` carries a `cg` dimension that selects this module in place of `wb4_slave_cdc` and drives the gating configuration; the same traffic, abort and post-abort phases run at several clock ratios in both directions.

11.7 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 – ch. 3 "WISHBONE Classic Bus Cycles", the protocol of the wrapped Wishbone block.
- Cummings, C.E. "Simulation and Synthesis Techniques for Asynchronous FIFO Design." *SNUG San Jose*, 2002 – the Gray-pointer async FIFO (`gaxi_fifo_async`) that carries the command and response across the two clocks.
- Ginosar, R. "Metastability and Synchronizers: A Tutorial." *IEEE Design & Test of Computers*, 28(5), Sept./Oct. 2011. DOI: 10.1109/MDT.2011.113 – synchronizer depth versus MTBF for the pointer crossings.
- Benini, L., Bogliolo, A., De Micheli, G. "A Survey of Design Techniques for System-Level Dynamic Power Management." *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 – the fixed-timeout policy `amba_clock_gate_ctrl` implements: gate after `cfg_cg_idle_count` idle cycles, wake on activity.
- Benini, L., De Micheli, G. "Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines." *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 – the latch-based gated clock (`icg`) at the bottom of the gating chain.

- Deschamps, J.-P., Sutter, G.D., Cantó, E. *Guide to FPGA Implementation of Arithmetic Functions*, Springer (LNEE 149), 2012, ch. 5 “Electronic Aspects of Digital Design”, §5.3.5.3 “Asynchronous FIFO” – textbook treatment (secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” – textbook treatment of clock gating (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `rtl/cdc/references/AsyncFifoGrayPointers__Cummings2002.pdf`, `rtl/cdc/references/MetastabilitySynchronizers__Ginosar2011.pdf`, `rtl/common/references/DynamicPowerManagementTimeout__Benini2000.pdf`, `rtl/common/references/LatchBasedClockGating__Benini1996.pdf`)

12 wb4_slave_cg

12.1 Overview

The **clock-gated variant** of wb4_slave: an `amba_clock_gate_ctrl` instance produces a gated clock that feeds an otherwise unmodified `wb4_slave`. Functionally identical to the base module; `cfg_cg_enable = 0` gives the base behaviour exactly. See the Shared book's Clock-Gated Variants Guide and amba_clock_gate_ctrl.

12.2 Parameters

In addition to all `wb4_slave` parameters:

Parameter	Type	Default	Description
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle countdown; bounds the programmable idle threshold

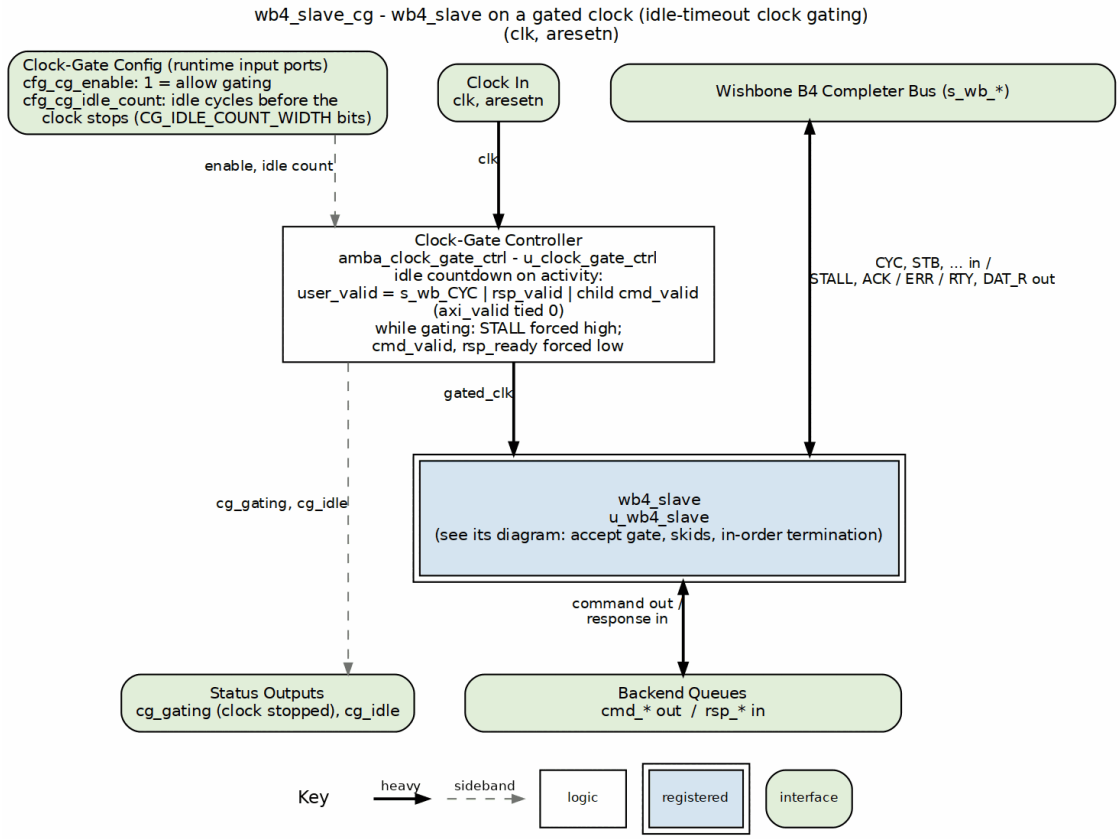
12.3 Ports

`wb4_slave`'s ports plus `cfg_cg_enable`, `cfg_cg_idle_count`, `cg_gating` and `cg_idle`, as on wb4_master_cg:

Port	Direction	Description
`cfg_cg_enable`	in	Global clock-gate enable
`cfg_cg_idle_count [CG_IDLE_COUNT_WIDTH-1:0]`	in	Idle clocks before the clock is gated
`cg_gating`	out	Clock is gated now
`cg_idle`	out	Nothing pending (the controller's idle indicator)

12.4 Functional Description

12.4.1 Block diagram



wb4_slave_cg block diagram

`wb4_slave` is one child box (see its diagram); this wrapper adds only the clock gate. The controller wakes on `s_wb_CYC`, `rsp_valid` or the child's `cmd_valid`. While the clock is stopped it holds STALL high on the bus and forces `cmd_valid` and `rsp_ready` low on the backend side.

12.4.2 Wake-Up Terms

Term	Why
`s_wb_CYC`	a bus cycle open: requests arriving, terminations due
`rsp_valid`	a FUB response offered
command pending on the output	a request waiting for the FUB

`cmd_ready` is deliberately absent.

12.4.3 Masks for the Wake-Latency Overlap

A stopped clock cannot register an accept, and the wake takes a clock or two after the activity appears. Three signals are therefore masked while `cg_gating` is high:

Signal	Masked to	Why
`cmd_valid` (to the FUB)	low	a consumer must not retire a command the slave cannot see leave
`rsp_ready` (to the FUB)	low	a FUB must not have a response taken by a stopped clock
`s_wb_STALL` (to the master)	**high**	in pipelined mode an accept is `STB && !STALL` in the master's clock; a frozen "room" would let the master move on from a request the slave never sampled

`STALL` stays high until the clock runs, and the master holds its request while stalled (a B4 rule), so nothing is lost and nothing is duplicated. Classic mode holds the request until termination and needs no `STALL`, but is masked the same way.

12.5 Design Notes

- Formal: `formal/amba/wb4_slave_cg/` proves the wrapper's glue contract with a clock-enable model of the gate cell (a derived clock is not provable in the repo's single-clock flow; the stopped clock itself is the cocotb test's job): reset state, gating clears by the third clock of an open cycle, no gating while a command is pending or with the enable low, no termination and no FUB handshake while gated, `STALL` high while gated; covers gating, an ungated accept, a termination and re-gating.
- The wake term is combinational into the controller (which registers it once), not a second local flop as in `apb4_slave_cg`; see `formal/amba/apb4_slave_cg/KNOWN_BUG.md` for the latency that flop adds.

12.6 Related Modules

- wb4_slave, wb4_master_cg

- apb4_slave_cg — the same wrapper over APB

12.7 Testing

`val/amba/test_wb4_slave_cg.py` runs the `wb4_slave` phases with gating enabled and the same three checks as the master's test.

12.8 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 "WISHBONE Classic Bus Cycles", the protocol of the wrapped Wishbone block.
- Benini, L., Bogliolo, A., De Micheli, G. "A Survey of Design Techniques for System-Level Dynamic Power Management." *IEEE Trans. VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 — the fixed-timeout policy `amba_clock_gate_ctrl` implements: gate after `cfg_cg_idle_count` idle cycles, wake on activity.
- Benini, L., De Micheli, G. "Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines." *IEEE Trans. CAD*, 15(6), June 1996. DOI: 10.1109/43.503933 — the latch-based gated clock (`icg`) at the bottom of the gating chain.
- Roy, S. *Advanced Digital System Design*, Springer, ch. 15 "Low-Power Digital System Design", §15.5.5 "Logic Level Power Down" — textbook treatment of clock gating (secondary source).
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`, `rtl/common/references/DynamicPowerManagementTimeout__Benini2000.pdf`, `rtl/common/references/LatchBasedClockGating__Benini1996.pdf`)

13 wb4_slave_stub

13.1 Overview

wb4_slave with its FUB-side queues presented as a single **packed vector** instead of named fields, matching what `apb4_slave_stub` and `apb5_slave_stub` offer. A shim-style converter that already carries a packed command through its own pipeline connects one bus instead of six, and the packing lives here rather than being re-derived by every consumer.

Nothing else differs: the stub instantiates the ordinary block and only splices the vector.

13.2 Parameters

Every parameter of wb4_slave, plus the derived widths:

Parameter	Value	Description
CPW	1 + AW + DW + SW + 3 + 2	Packed command width
RPW	2 + DW	Packed response width

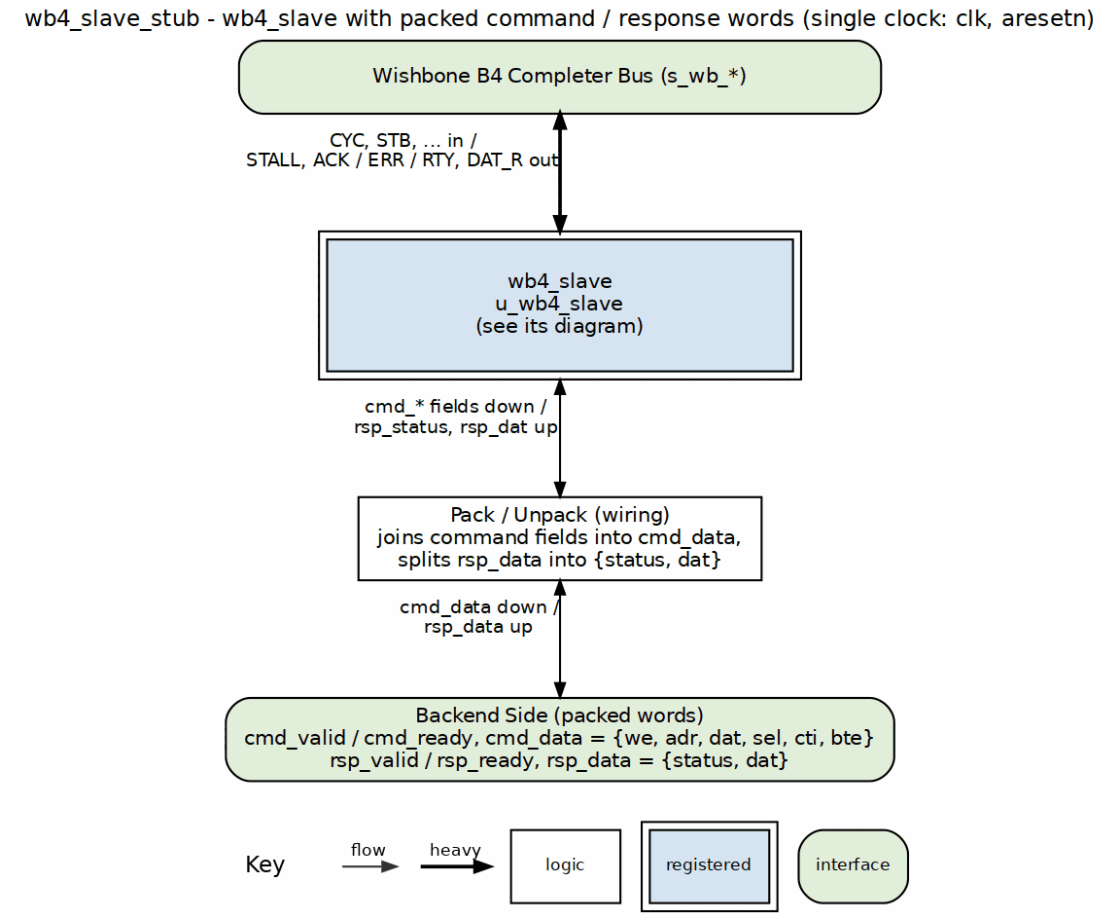
13.3 Ports

`wb4_slave`'s Wishbone side, unchanged. On the FUB side, the command leaves packed and the response arrives packed:

Port	Direction	Description
`cmd_valid` / `cmd_ready`	Output / Input	Command handshake
`cmd_data [CPW-1:0]`	Output	Packed command
`rsp_valid` / `rsp_ready`	Input / Output	Response handshake
`rsp_data [RPW-1:0]`	Input	Packed response

13.4 Functional Description

13.4.1 Block diagram



wb4_slave_stub block diagram

A packing wrapper: `wb4_slave` is one child box (see its diagram). The pack/unpack stage is wiring only: it joins the command fields into `cmd_data` for the backend and splits `rsp_data` back into `{status, dat}`.

13.4.2 Packing

Most significant field first. Both stubs use the same order, so a `wb4_master_stub` and a `wb4_slave_stub` connect directly:

```
cmd_data = {we, adr, dat, sel, cti, bte}      // CPW = 1 + AW + DW + SW
+ 3 + 2
rsp_data = {status, dat}                      // RPW = 2 + DW
```

`cti` and `bte` hold their bits whatever `USE_BURST_HINTS` is, so the vector width does not move with the parameter. With the hints off the inner block ignores them and the bus reads CLASSIC/LINEAR.

13.5 Related Modules

- wb4_slave — the block this wraps
- wb4_master_stub — the other half of a stub pair
- wb4_pkg — the status encoding inside `rsp_data`

13.6 Testing

`val/amba/test_wb4_stubs.py` checks the packing directly, in both directions and with the burst hints on and off. Two mutations fail it: swapping `dat`/`sel` in the master's unpack, and reversing the slave's response unpack.

13.7 References

- Herveille, R. (ed.). *Wishbone B4: WISHBONE System-on-Chip (SoC) Interconnection Architecture for Portable IP Cores*. OpenCores, 2010 — ch. 3 "WISHBONE Classic Bus Cycles", the protocol of the wrapped Wishbone block.
- Indexed in `rtl/amba/references/README.md` (local PDF: `WishboneB4Spec__OpenCores2010.pdf`)